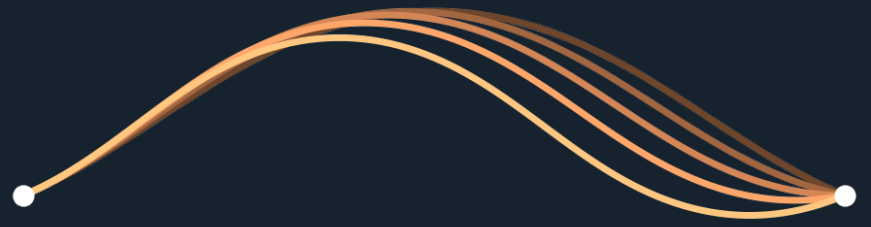
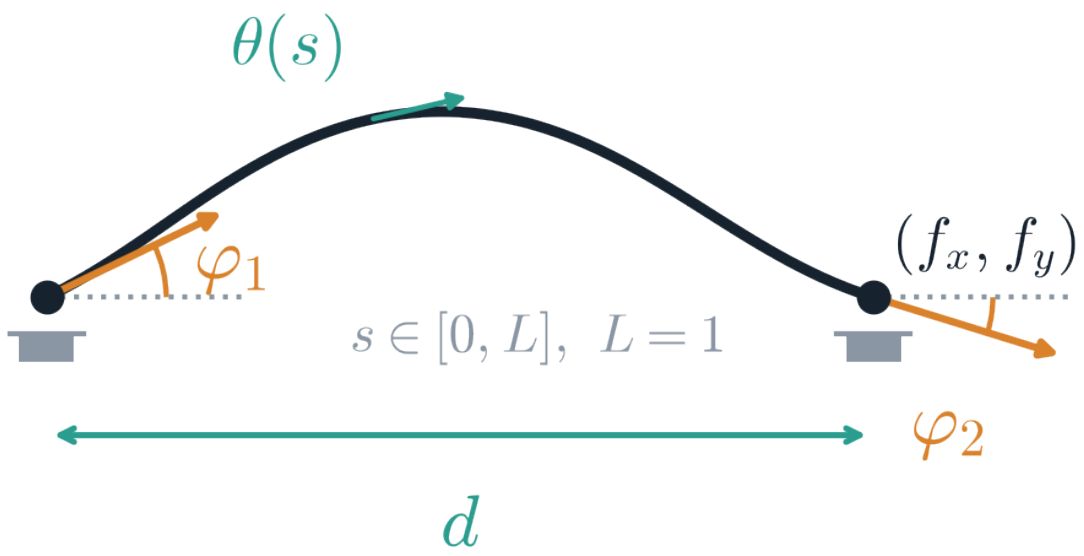
Mapping the clamped elastica across its snap surface

Continuation with buckling-mode branch tracking in FEniCSx / PETSc



One rod, three parameters, a map of equilibria



For every point (d, φ_1, φ_2) solve for the tangent angle θ(s) and the end forces f_x, f_y; record both clamp moments and the bending energy U.

d

End-to-end distance (L = 1)

0.70 – 0.99; Δd = 0.01

φ_1

Clamp angle at s = 0

−π/2 to π/2; Δφ = π/314 ≈ 0.01

φ_2

Clamp angle at s = L

Same range as φ_1; grid 30 × 315 × 315

Equilibria are stationary points of the bending energy

Bending energy

Inextensible Euler–Bernoulli rod, EI = 1

$$\Pi[\theta] = \frac{1}{2}\int_0^L \theta'(s)^2\, ds$$

Constraints

End position fixed, both ends clamped

$$\int_0^L \cos\theta\ ds = d, \quad \int_0^L \sin\theta\ ds = 0, \quad \theta(0) = \varphi_1, \quad \theta(L) = \varphi_2$$

Lagrangian

f_x, f_y enforce the two end-position integrals

$$\mathcal{L}[\theta, f_x, f_y] = \frac{1}{2}\int_0^L \theta'^2\, ds + f_x\left(\int_0^L \cos\theta\ ds - d\right) + f_y \int_0^L \sin\theta\ ds$$

f_x and f_y are the Lagrange multipliers of the end-position constraints — physically, the reaction forces at the right clamp. The study maps every stable stationary point of $\mathscr{L}$ the continuation reaches over the (d, φ_1, φ_2) box.

Stationarity gives the elastica equation and a mixed weak form

Euler–Lagrange equation

$$\theta'' + f_x \sin\theta - f_y \cos\theta = 0$$

$\theta(0) = \varphi_1,\ \theta(L) = \varphi_2$

1. $\theta \in$ P1 Lagrange on 200 elements, graded towards both clamps
2. $f_x, f_y \in$ “Real” space: one global DOF each
3. Clamp angles imposed as Dirichlet BCs on θ

Residuals assembled by FEniCSx (solver.py)

$$F_0 = \int_0^L \left(\theta' v' - f_x \sin\theta\ v + f_y \cos\theta\ v\right) ds = 0 \quad \forall v \in V_0$$

$$F_1 = \int_0^L (\cos\theta - d)\, q_x\, ds = 0$$

$$F_2 = \int_0^L \sin\theta\ q_y\, ds = 0$$

Post-processed per point

$$M_1 = \theta'(0), \quad M_2 = \theta'(L), \quad U = \frac{1}{2}\int_0^L \theta'^2\, ds$$

Moments: end-element slope plus a half-element Taylor correction (second order)

Each parameter point is one Newton solve on 203 unknowns

$$J(\mathbf{u}^k)\,\delta\mathbf{u} = -\mathbf{F}(\mathbf{u}^k), \qquad \mathbf{u}^{k+1} = \mathbf{u}^k + \alpha_k\,\delta\mathbf{u}$$

u = ($\theta_0 \ldots \theta_{200}$, f_x, f_y); ($0 < \alpha_k < 1$) from cubic backtracking line search

203

unknowns per point

201 nodal angles + 2 multipliers

LU

MUMPS direct solve

KSP preonly: exact linear steps

10^{-10}

relative tolerance

absolute 10^{-12}

50

max Newton its

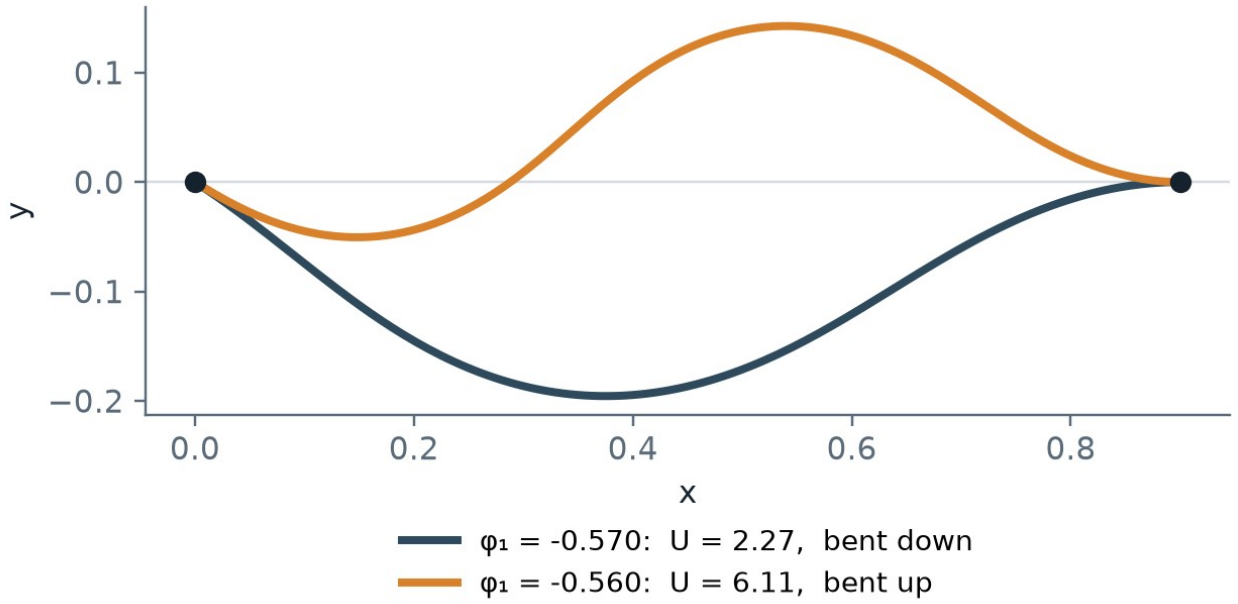
per warm start; more counts as a failure

At a fold of the solution branch, the Jacobian becomes singular, so the snap surface is exactly where the Newton solver fails, and where the continuation logic has to take over.

The snap surface

Where the equilibrium jumps from one branch to another

Two equilibria 0.01 rad apart: the rod has snapped



d = 0.90, φ_2 = 0, production run. Converged solutions on either side of the jump.

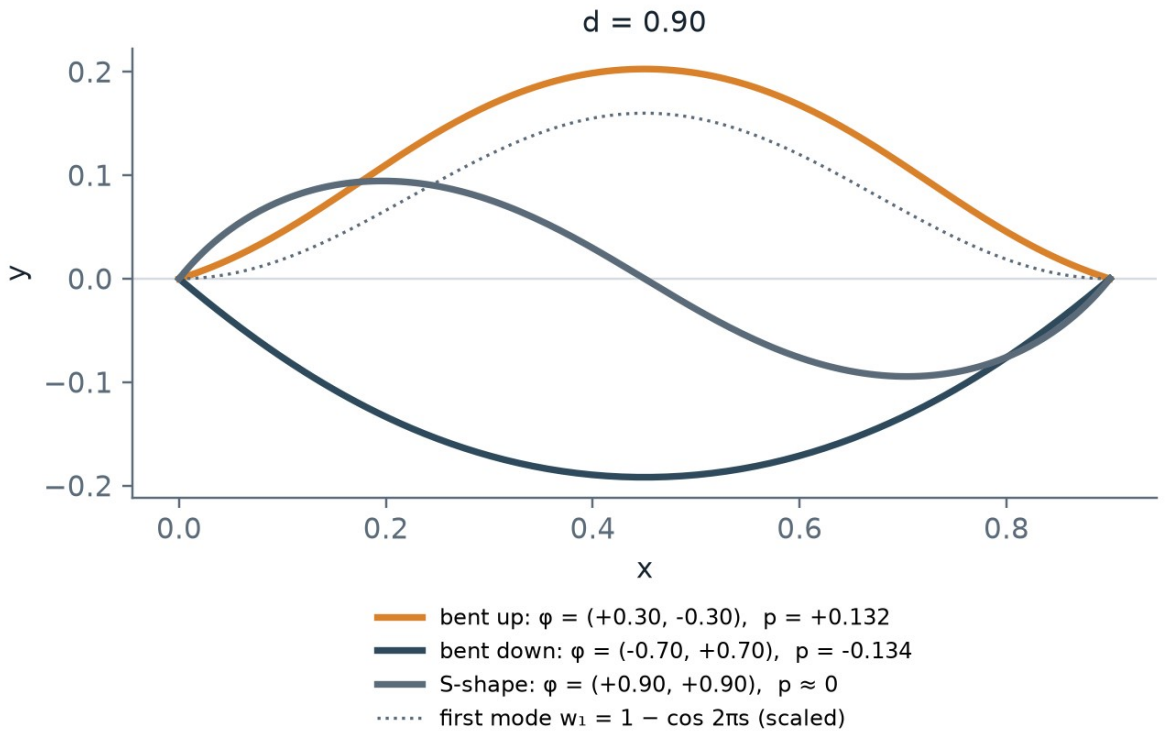
Why Newton struggles here

1. **Fold**
   The branch ends; the Jacobian goes singular and Newton steps blow up.

2. **Silent branch switch**
   A warm start from one side converges, but to the other branch.

3. **Residual can't tell**
   Both sides meet 10^{-10}, so convergence alone says nothing about which branch you are on.

Bent up or bent down: the first buckling mode decides



$$p = \frac{1}{\|w_1\|}\int_0^L y\, w_1\, ds$$

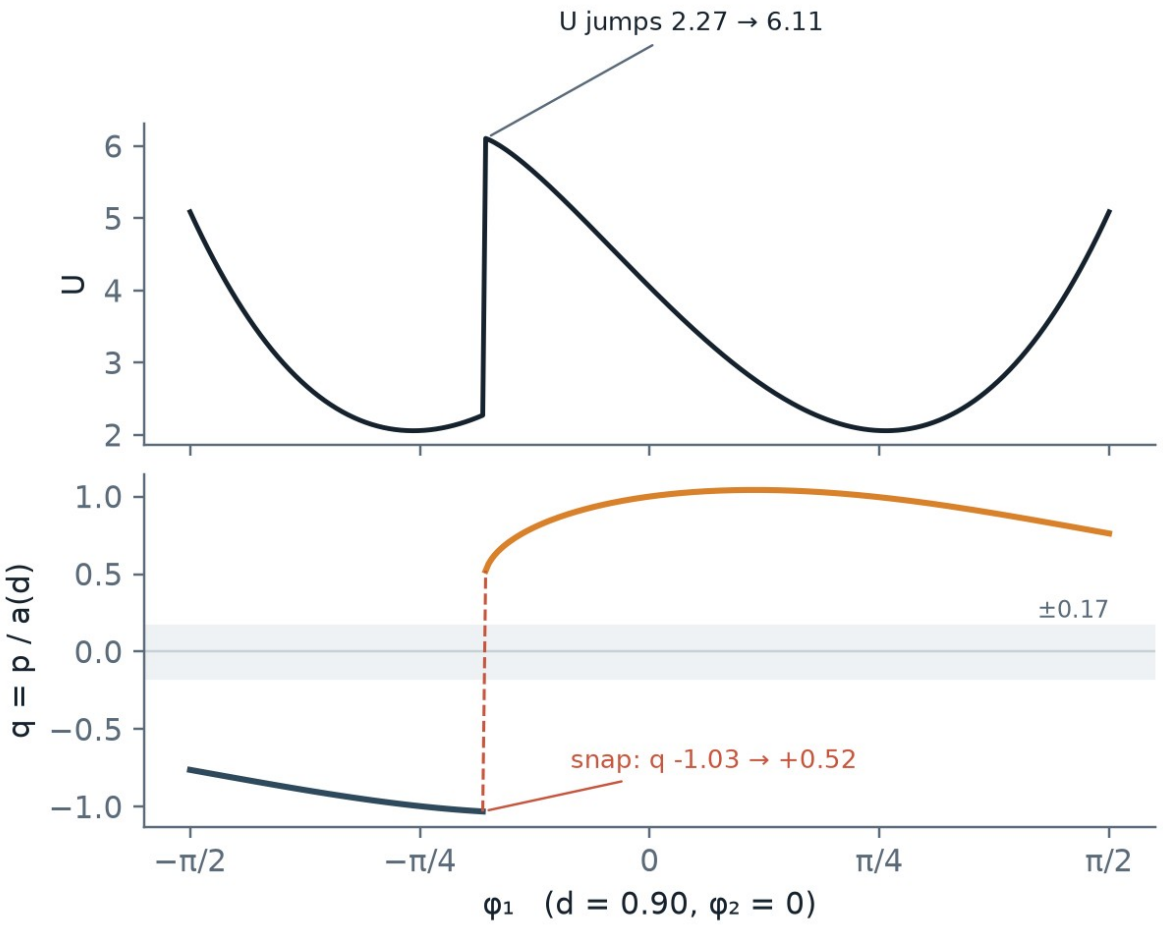
$$y(s) = \int_0^s \sin\theta\, dt, \qquad w_1(s) = 1 - \cos\frac{2\pi s}{L}$$

1 **First clamped buckling mode**
w_1 is the symmetric hump; the second mode is antisymmetric and carries no up/down information.

2 **Sign = branch**
$p > 0$: bent up. $p < 0$: bent down. The S-shape has $p \approx 0$.

3 **Built-in symmetry**
p is unchanged by the mirror $(\varphi_1, \varphi_2) \to (-\varphi_2, -\varphi_1)$ and flips under $(\varphi_1, \varphi_2) \to (-\varphi_1, -\varphi_2)$.

A snap is a sign change of p with a jump



$$q = \frac{p}{a(d)}$$

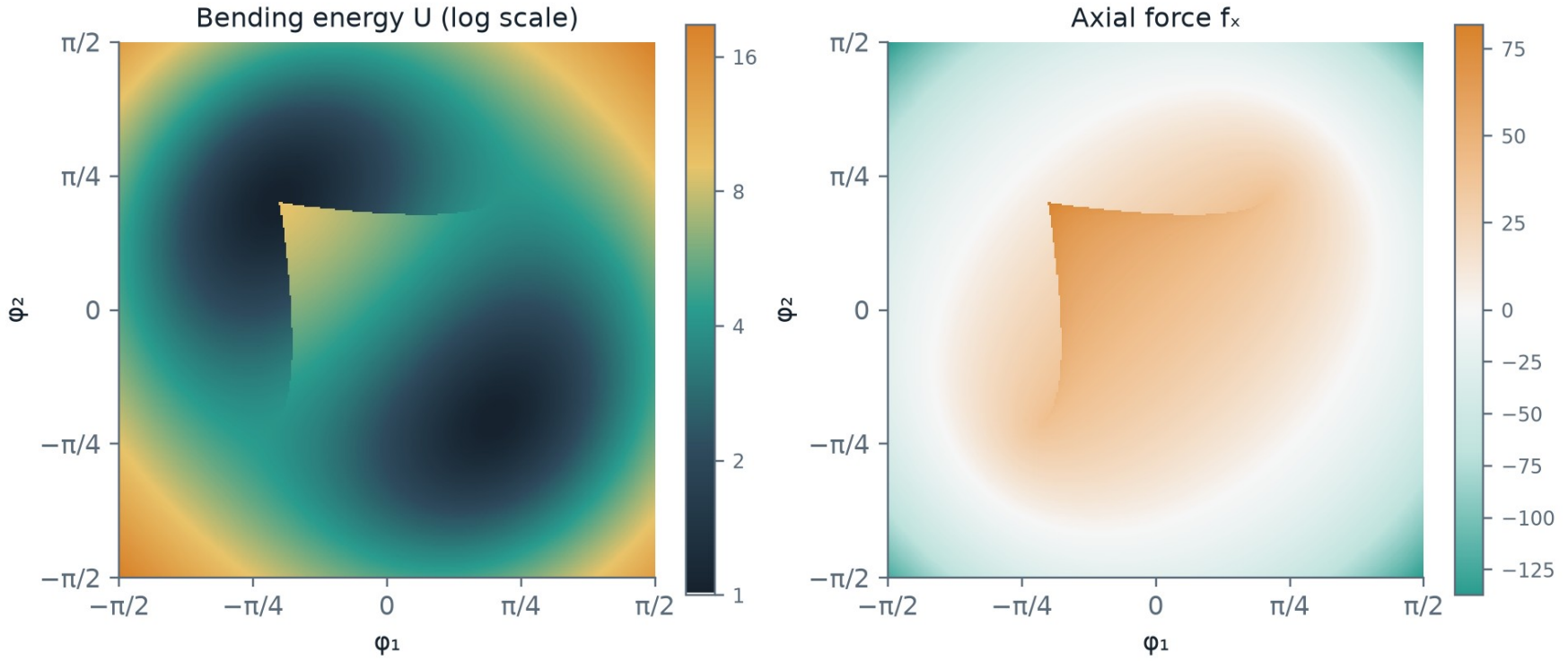
$$|\Delta q| > 0.17 \quad \text{and} \quad |\Delta q| > \frac{1}{2}\max(|q_a|, |q_b|)$$

1. **Scale out d**
   p shrinks like √(1 − d); a(d) is |p| of the exact first mode at (d, 0, 0), so q = ±1 there at every d.

2. **Snap**
   Sign change with a jump: q goes −1.03 → +0.52 in one 0.01 step. Smooth neighbours differ by about 0.003.

3. **Smooth crossing**
   Sign change without a jump, through the S-shape near $\varphi_1 = \varphi_2$. Not a snap.

Smooth fields, then a jump across an L-shaped surface



Production run, $d = 0.90$ layer: $\varphi_1, \varphi_2 \in [-\pi/2, \pi/2]$, $\Delta\varphi \approx 0.01$, 99,225 points.

Smooth interior

Grid neighbours differ by $|\Delta q| \approx 0.003$

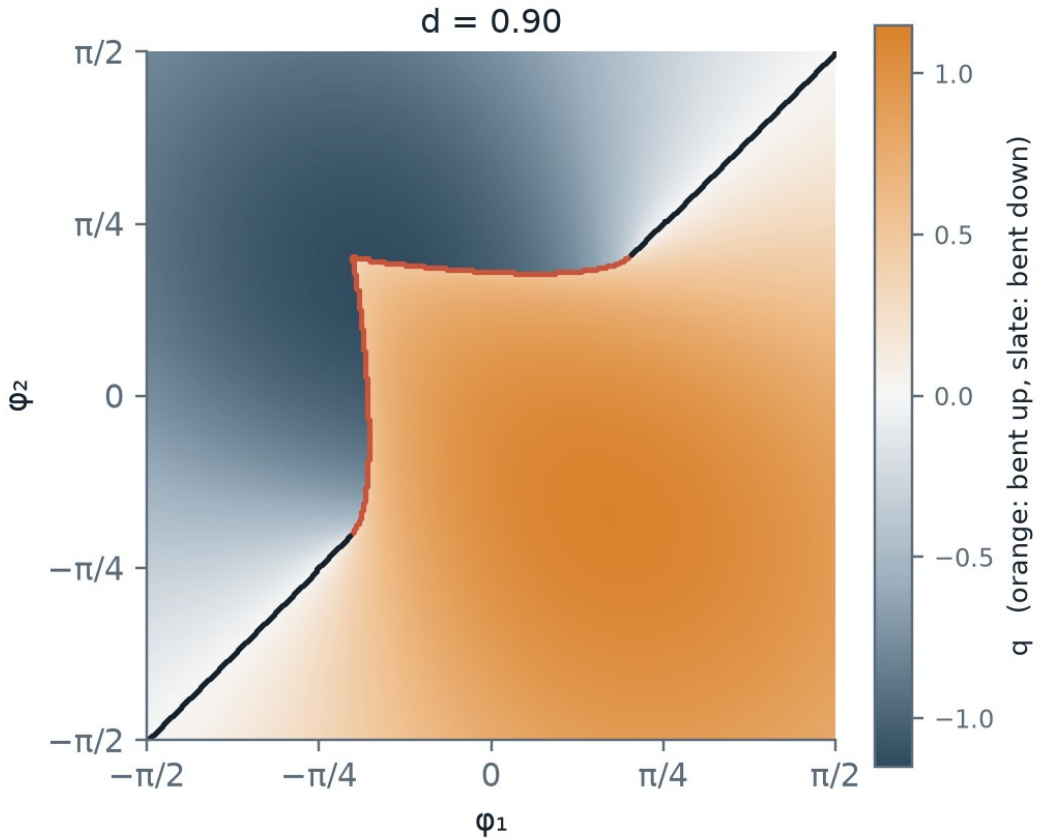
Sharp edge

Jump at $\varphi_1 \approx -0.57$ and, by symmetry, $\varphi_2 \approx +0.57$

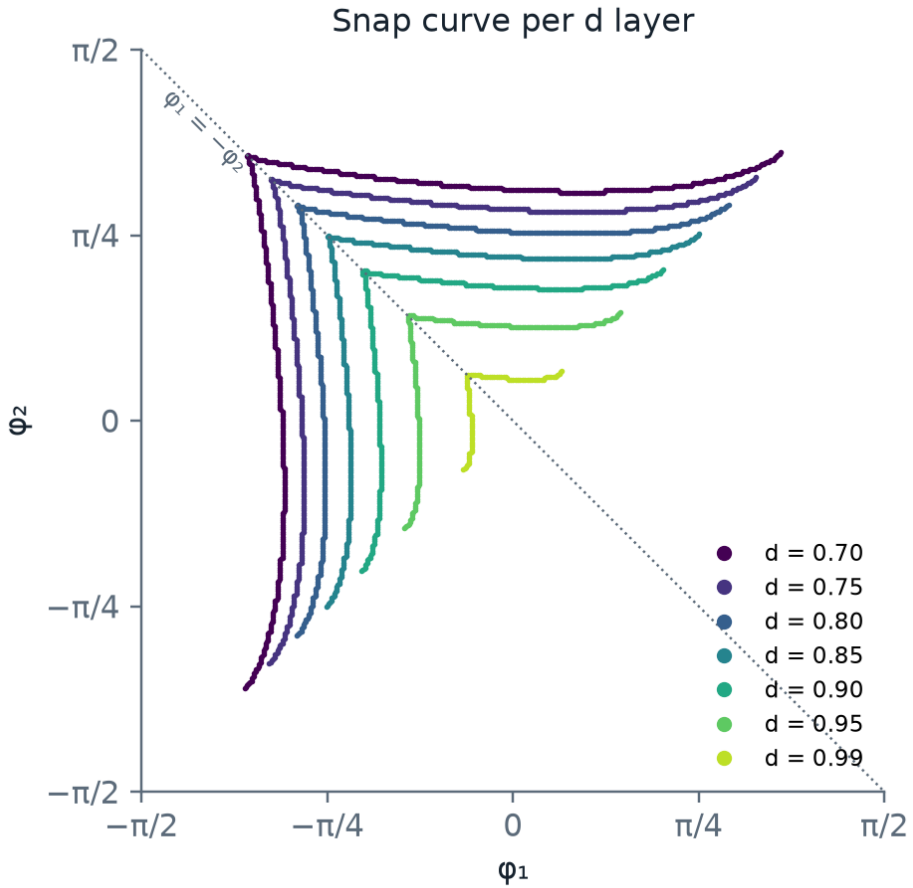
Large jump

U 2.27 → 6.11 and f_x 28 → 59 in a single 0.01 step

The sign of p traces the snap surface

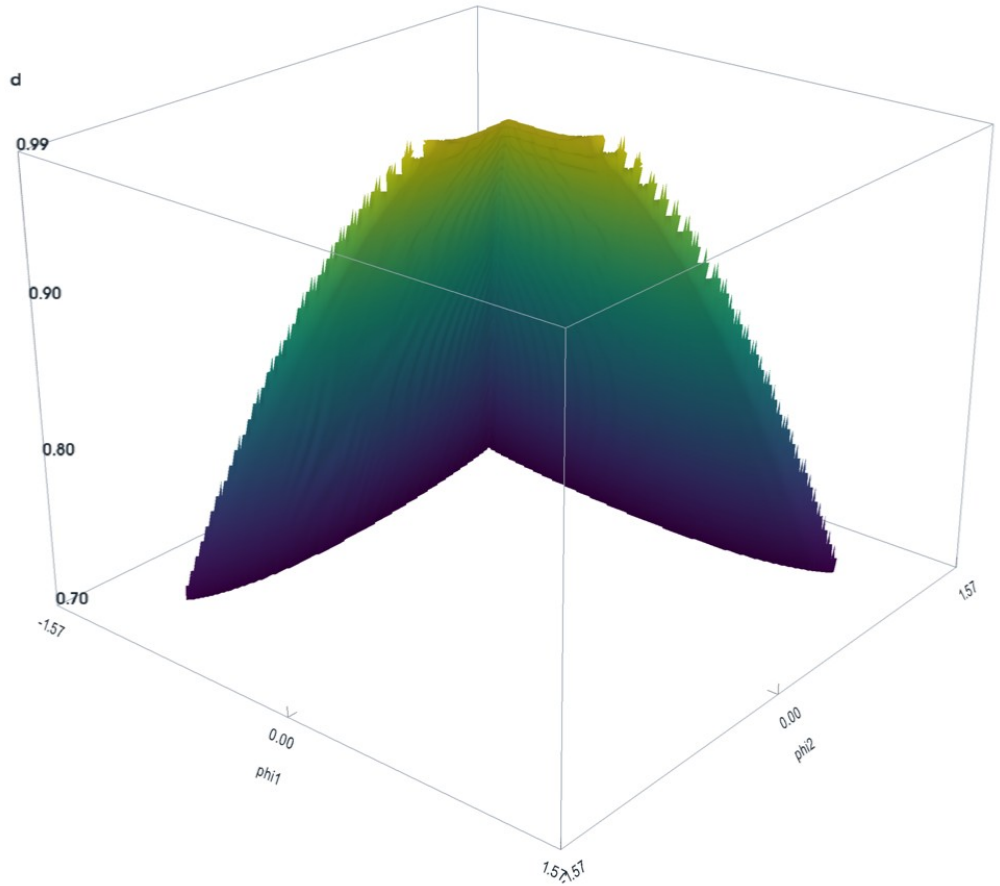


Red: snap edges, where p changes sign with a jump. Black: smooth crossings through the S-shape.

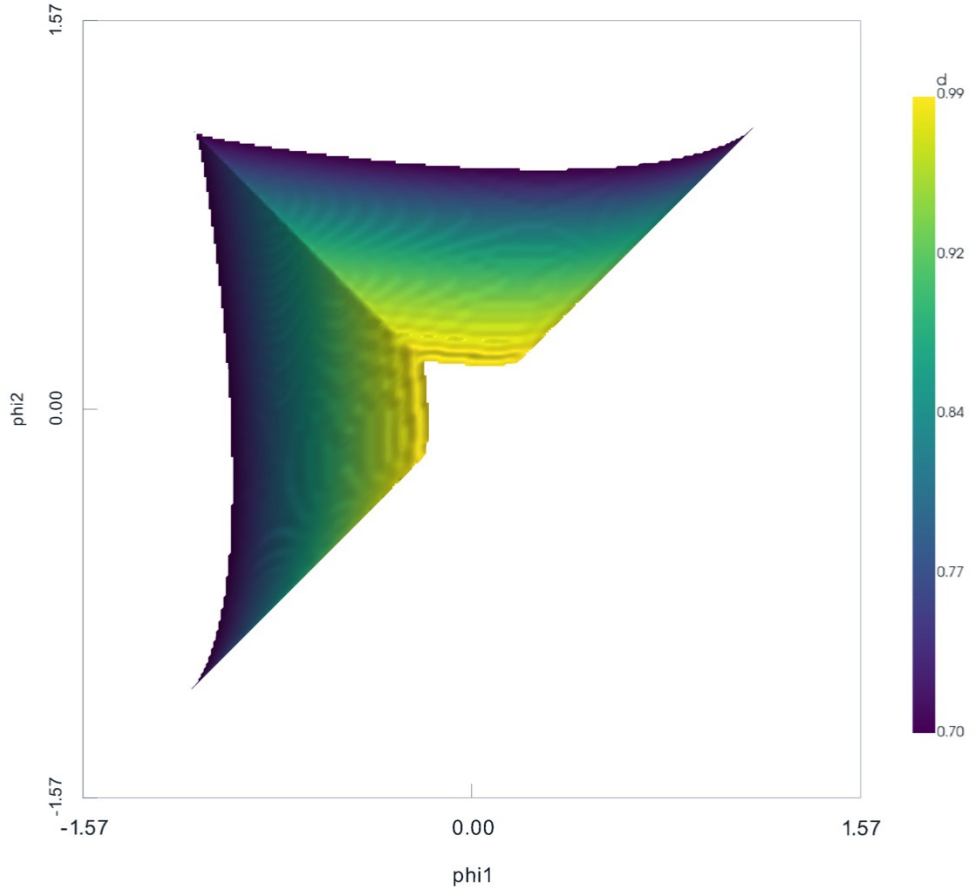


As d → 1 the L closes in: at $\varphi_2 = 0$ the snap sits at $\varphi_1 \approx -0.98$ (d = 0.70), −0.57 (0.90) and −0.18 (0.99).

The snap surface in 3D

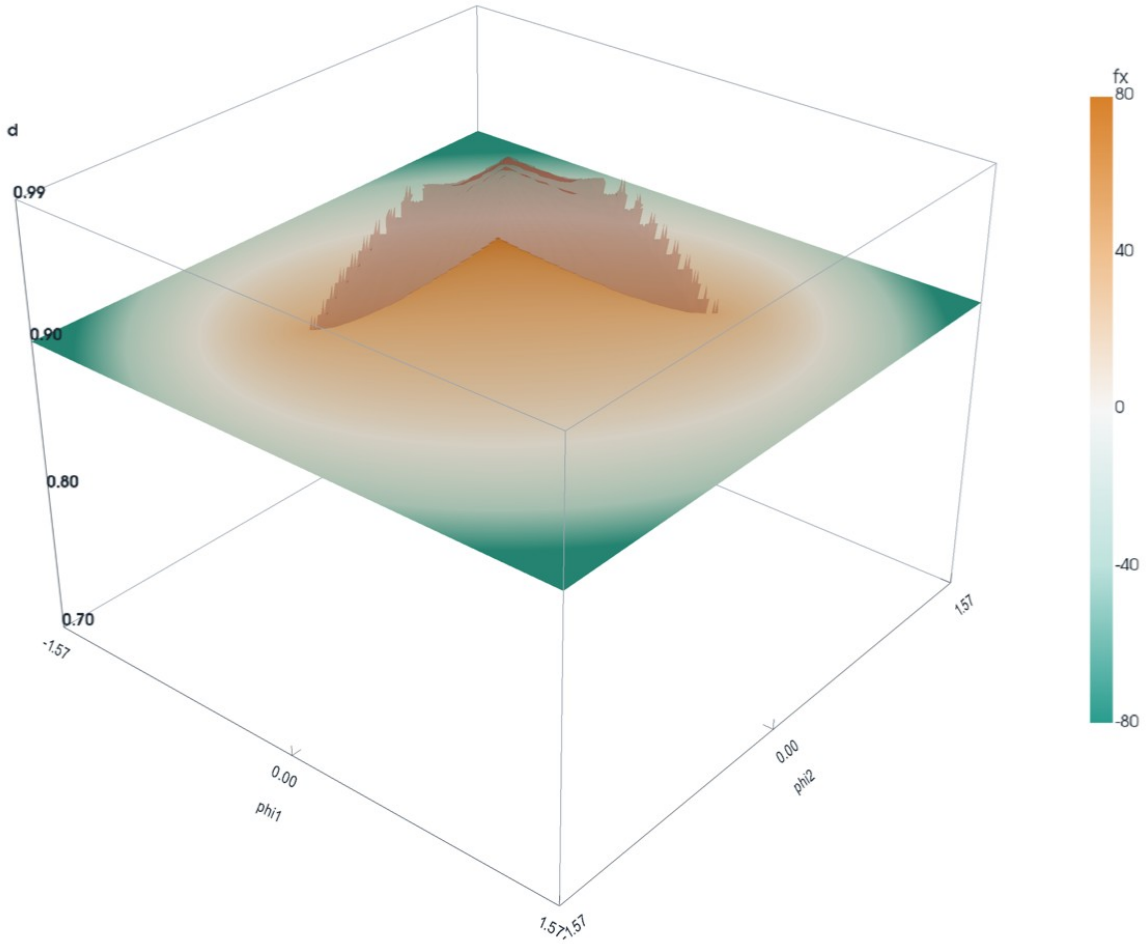


Oblique view, d stretched ×8. The p = 0 contour of the 17.2 M-tetrahedron grid, kept where p jumps: 30,081 snap edges, area 2.64.



Down the d axis: every layer is the L-shape, and the sheet closes in on the origin as d → 1. Colour: d.

The sheet rises from the step in axial force



Layer d = 0.90 coloured by f_x (clipped at ±80); the snap sheet above it is translucent.

Across the snap at $\varphi_2 = 0$, f_x roughly doubles at every d:

30.1 → 61.5 — d = 0.80, jump at $\varphi_1 \approx -0.80$

28.3 → 58.7 — d = 0.90, jump at $\varphi_1 \approx -0.57$

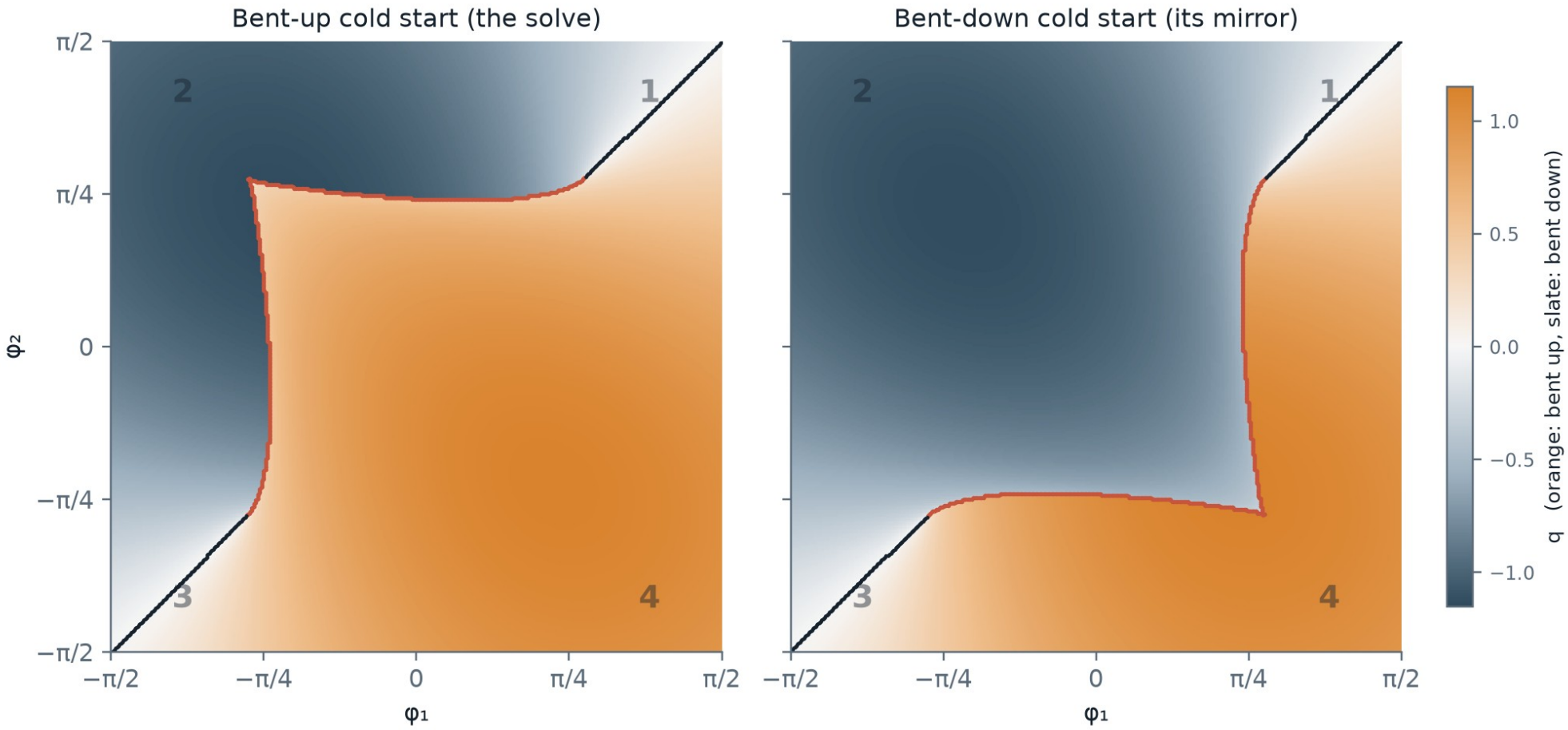
27.4 → 55.4 — d = 0.97, jump at $\varphi_1 \approx -0.31$

The surface is built from the shape alone, through the sign of p, yet it lands exactly on the step the forces see.

The bistable region

Two stable rods for the same clamps, and which snap the rod can meet

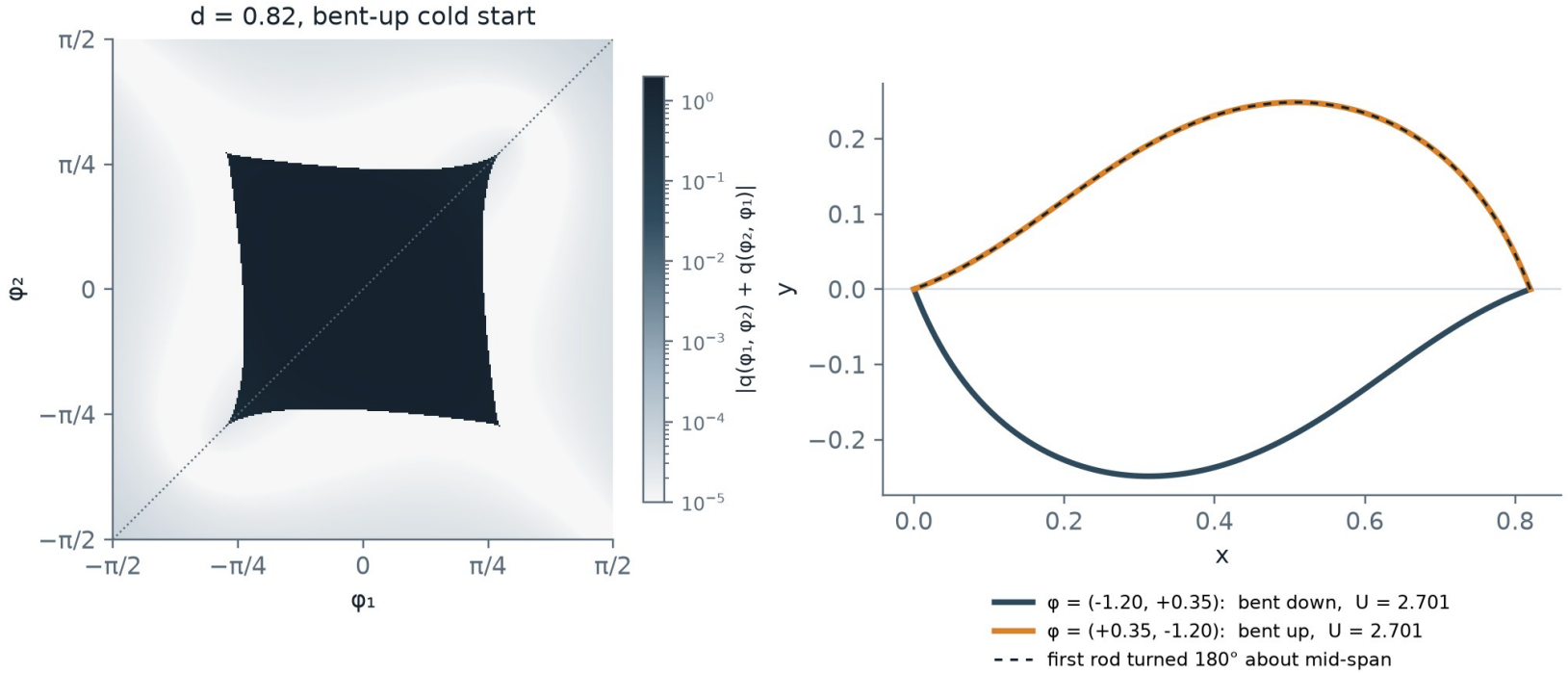
A bent-down cold start mirrors the map across $\varphi_1 = \varphi_2$



d = 0.82. θ → −θ takes the solve to the bent-down cold start: $(\varphi_1, \varphi_2) \to (-\varphi_1, -\varphi_2)$ with p negated. With the mirror $(\varphi_1, \varphi_2) \to (-\varphi_2, -\varphi_1)$ that is a reflection in $\varphi_1 = \varphi_2$ with orange and slate swapped. Red: each map's snap curve, in quadrants 1, 2, 3 and 1, 3, 4.

Outside the region, swapping the clamps turns the rod over

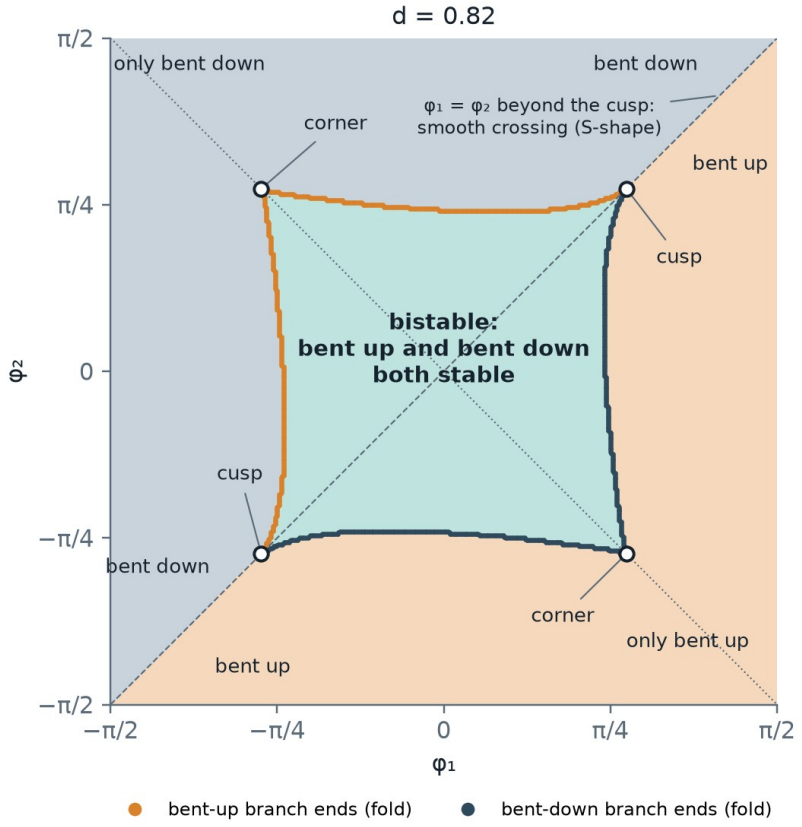
$$(d, \varphi_1, \varphi_2) \mapsto (d, \varphi_2, \varphi_1): \quad \tilde{\theta}(s) = \theta(L - s), \quad \tilde{p} = -p, \quad \tilde{U} = U, \quad \tilde{f}_x = f_x, \quad \tilde{f}_y = f_y, \quad \tilde{M}_1 = -M_2$$

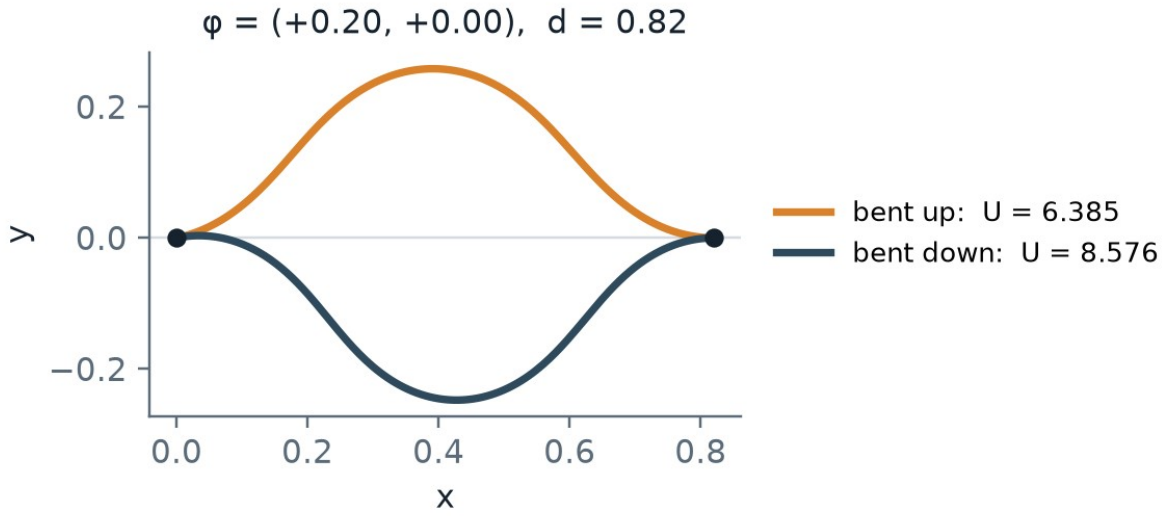


Left: $|q(\varphi_1, \varphi_2) + q(\varphi_2, \varphi_1)|$ at d = 0.82 is about 1e-4 everywhere except a square, the bistable region. Over all layers 2.35 M of the 2.98 M nodes pass, with q within 6.6e-4 and U within 1.5e-8. Right: the rod at (−1.20, +0.35) turned 180° lies on the rod at (+0.35, −1.20).

The bistable region: two stable rods for the same clamps

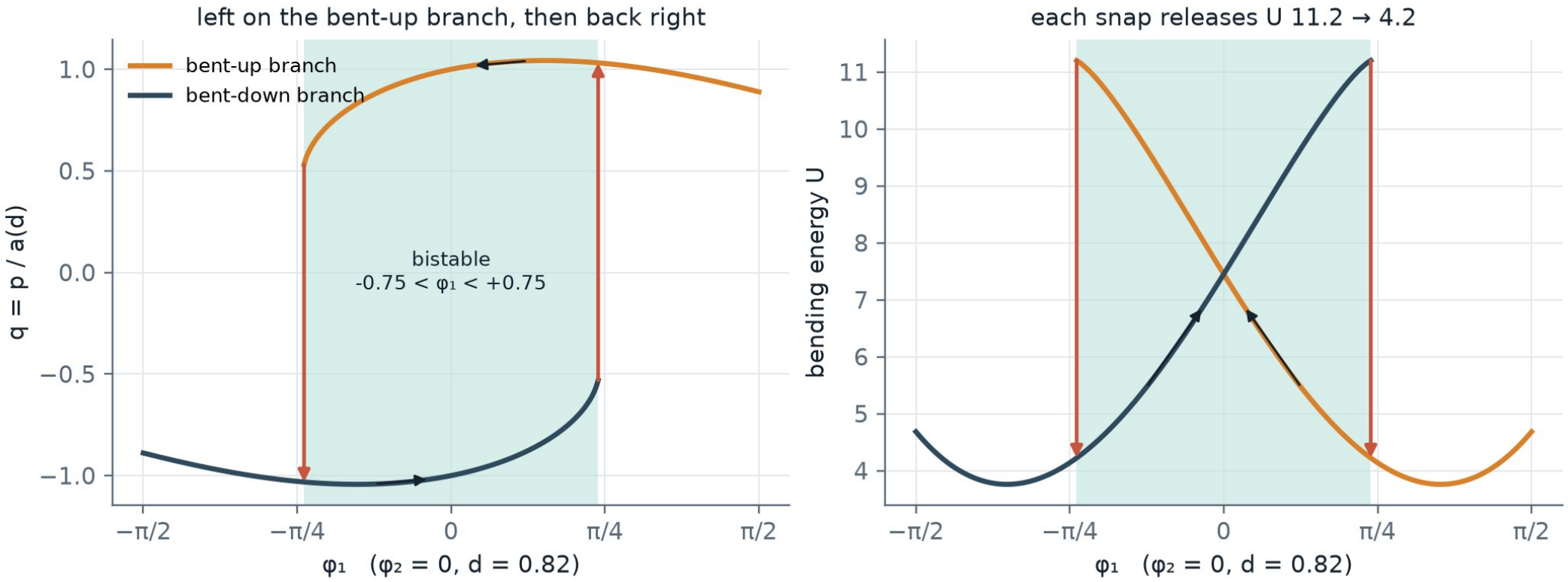
Bistable where the bent-up solve is bent up at a point and at its mirror: $q(\varphi_1, \varphi_2) > 0$ and $q(\varphi_2, \varphi_1) > 0$





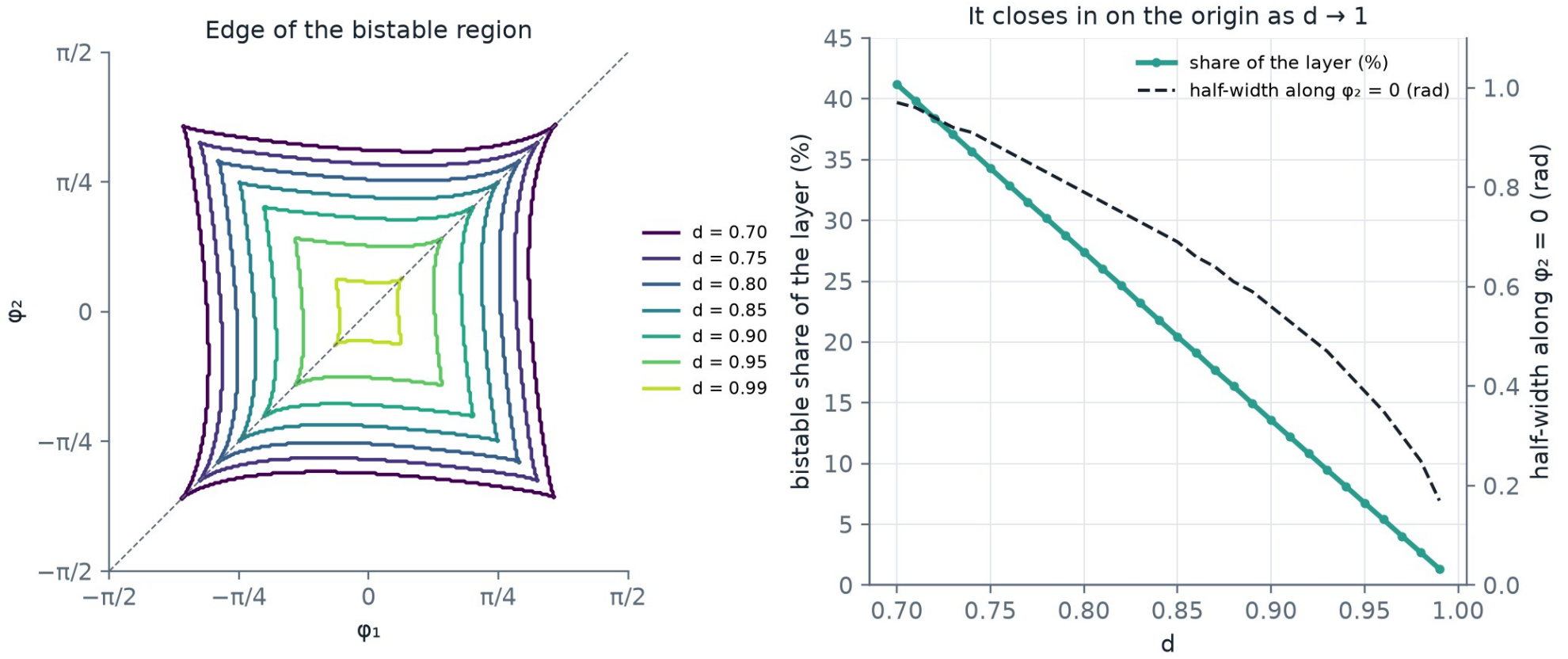
1. Left and top: the bent-up branch ends (fold). Leave this way bent up and the rod snaps down.
2. Bottom and right: the bent-down branch ends, the mirror image. Leave this way bent down and the rod snaps up.
3. On the diagonal the curves meet at cusps. Beyond them $\varphi_1 = \varphi_2$ is a smooth S-crossing and nothing snaps.

Which snap the rod meets depends on the branch it is on



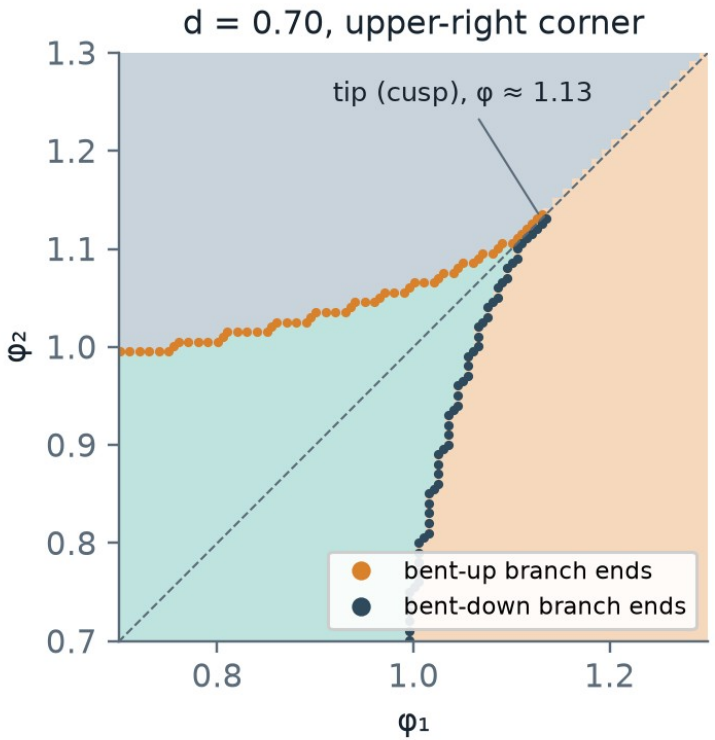
d = 0.82, $\varphi_2 = 0$. Bent up (orange), the rod crosses the whole bistable band (teal) and snaps down only at $\varphi_1 \approx -0.76$. Bent down (slate), it snaps up only at +0.76. The loop between the two snaps is the hysteresis; each snap releases U 11.2 → 4.2.

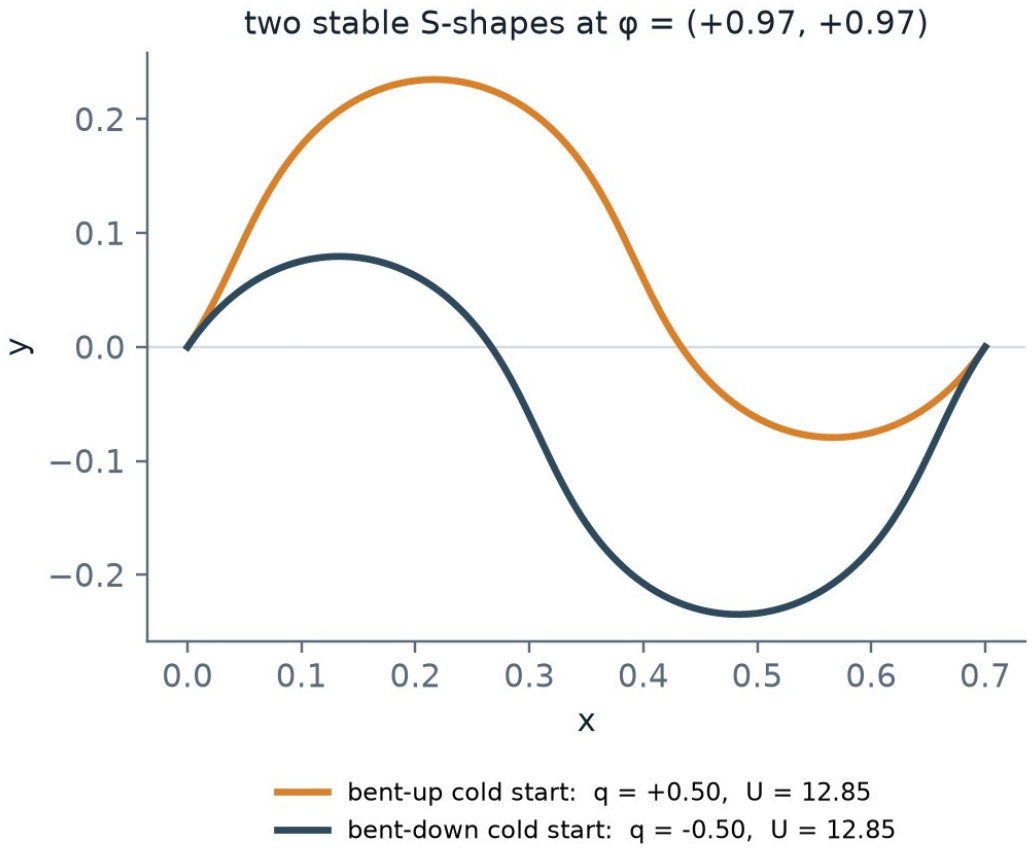
The bistable region closes in as d → 1



Share of the layer: 41 % at d = 0.70, 25 % at 0.82, 14 % at 0.90, 1.3 % at 0.99. Along $\varphi_2 = 0$ it spans $|\varphi_1| < 0.97$, 0.75, 0.56 and 0.17. The largest energy gap between the two rods falls from 25.6 to 0.8.

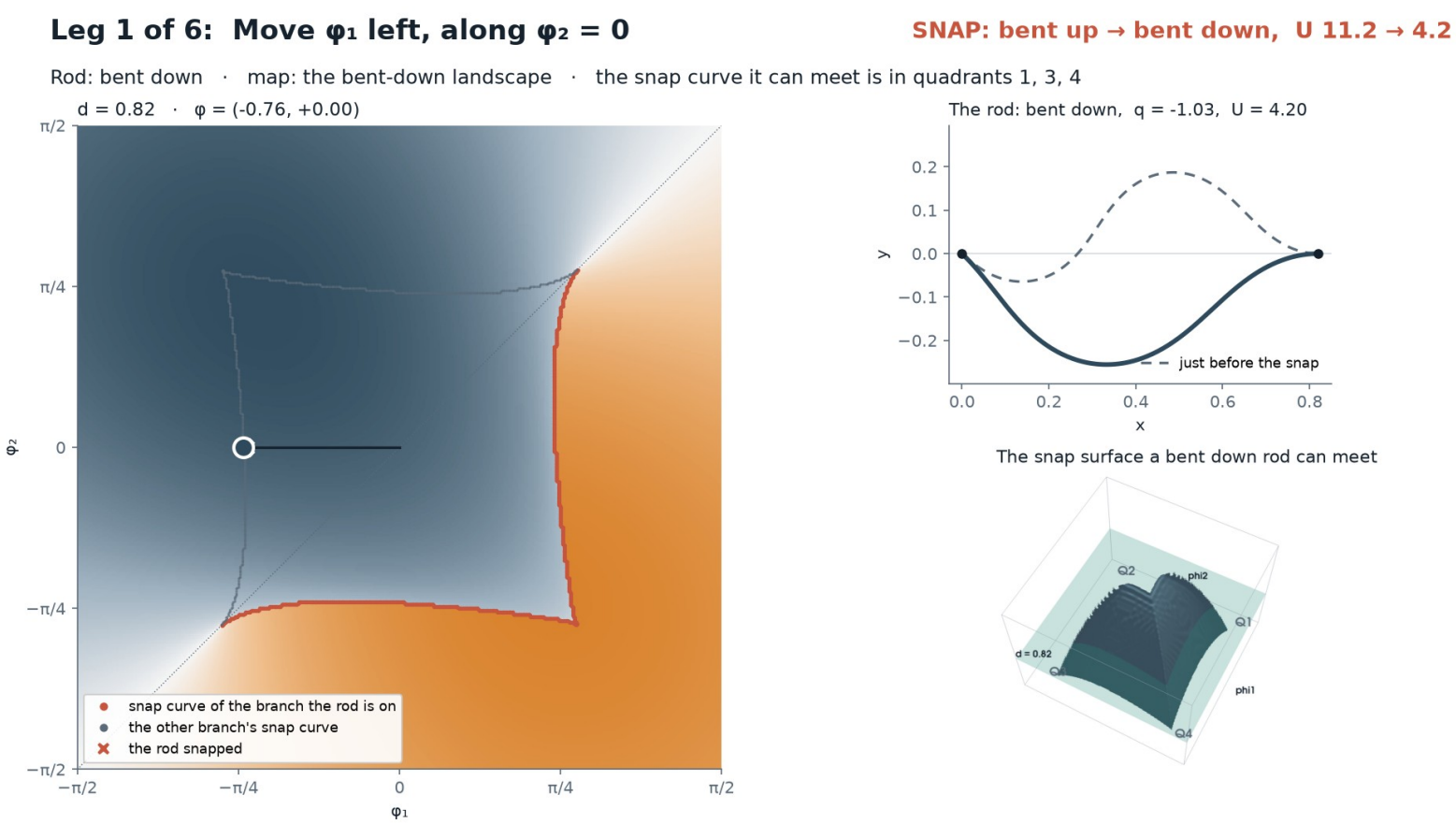
The region ends in a cusp on the diagonal





d = 0.70. Towards the diagonal the two fold curves close in and meet at a cusp at φ ≈ 1.13. Near it both stable rods are S-shapes, mirror images with q ≈ ±0.5 and equal energy on the diagonal; at the cusp q falls to 0. The cusp moves in with d: φ ≈ 0.86 at d = 0.82, 0.20 at 0.99.

Following one rod around the snap surface



Click to play (29 s). d = 0.82. Left: the map of the branch the rod is on; red, the snap curve it can meet; grey, the other branch's. Right: the rod, and the snap surface in (φ₁, φ₂, d) for its branch.

1. Across. Bent up, the rod snaps down at $\varphi_1 = -0.76$, and the map and snap surface switch to the bent-down ones. Coming back it snaps up only at +0.76.

2. Around. Raising φ_2 at $\varphi_1 = 1.25$ passes the end of the snap curve: a smooth S-crossing turns the rod bent down, no snap.

3. Back at the start, now bent down. Repeating the first move, nothing snaps: its snap curve is in quadrants 1, 3, 4.

Continuation strategy

How the solver walks the parameter space and guards against branch switches

Traverse the grid; reject jumps and unstable results

Cold-start once at $\varphi_1 = \varphi_2 = 0$, smallest d; visit nodes by grid distance from it

→

Warm-start θ, f_x, f_y from a solved grid neighbour: bent-up ones first, then nearest

→

Newton solve (SNES, at most 50 iterations)

→

No jump in q, and stable?

yes →

Accept, store, and write the mirror point

no ↓

Rejected or diverged: try the next neighbour. After 3 failures from one branch, wait for the other; a waiting node retries whenever a neighbour is solved.

Grid

Rectilinear (d, φ_1, φ_2) grid split into 6 tetrahedra per cell; warm starts come from the 26 grid neighbours

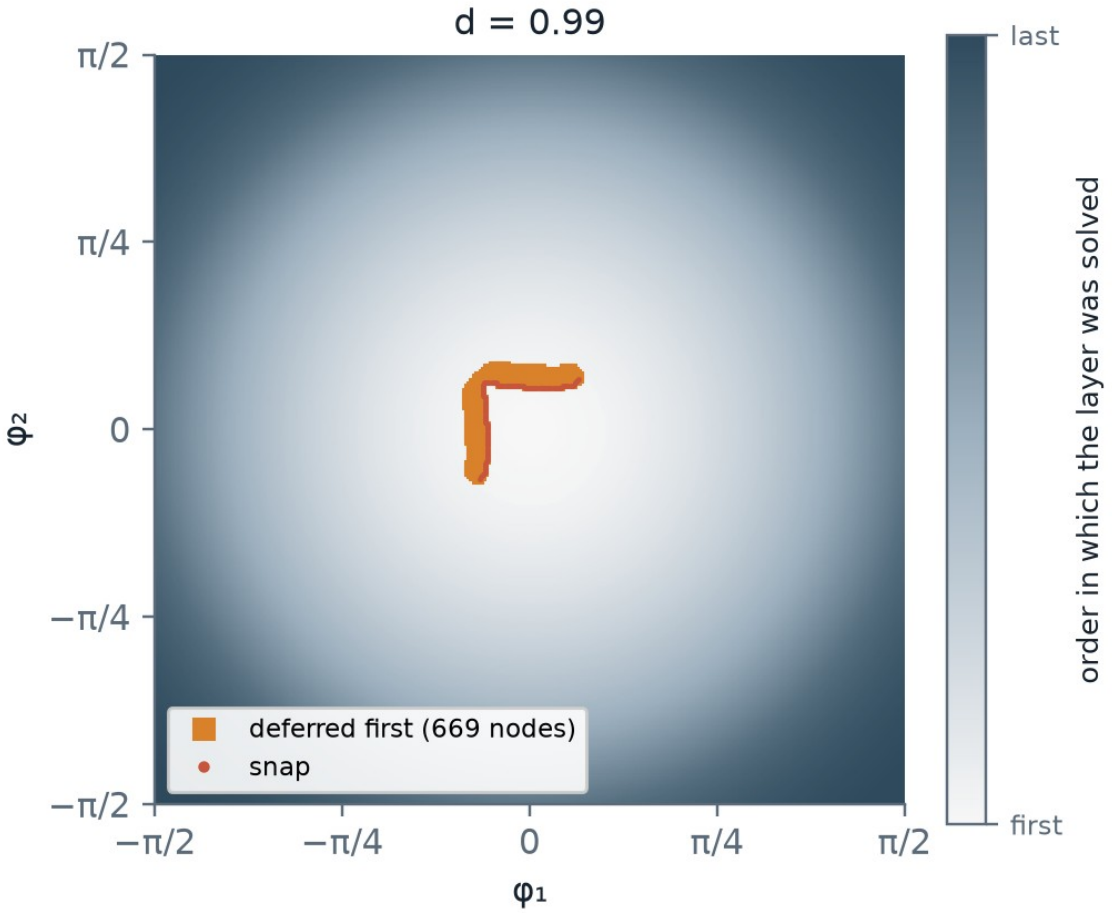
Restartable

Live NPZ checkpoints, also on Ctrl-C; a resume reloads every stored θ as a warm-start source

Half domain

Only $\varphi_1 + \varphi_2 \geq 0$ is solved; a precomputed mirror index fills the other half

Going around a fold: defer, never jump



d = 0.99 layer: the front grows outward; orange nodes had to wait for a neighbour on the other branch.

1. **Defer**
   A node with no acceptable warm start is set aside, not forced; it is retried each time a new neighbour is solved.

2. **Wait for the other branch**
   After 3 failures from neighbours on one branch it only tries the other, so a node past a fold waits for the far side.

3. **The long way round**
   The front reaches the cut-off region through the smooth crossings and approaches it from the far side.

4. **Last resort**
   Only if the whole front stalls is the nearest waiting node cold-started.

Production run: 1,155 nodes waited (all at d ≥ 0.97); one extra cold start, at (0.99, 0.20, 0.20). A final promotion pass gave the bent-up rod to 1,571 bistable nodes the front had filled bent down.

A stability test catches what the jump test misses

Constrained second variation

$$Q[v] = \int_0^L v'^2 - (f_x \cos\theta + f_y \sin\theta)\, v^2 \, ds$$

$$v(0) = v(L) = 0, \qquad \int_0^L v\cos\theta \, ds = \int_0^L v\sin\theta \, ds = 0$$

$$n_- = \mathrm{neg}(A) + \mathrm{pos}(C^T A^{-1} C) - 2$$

Stable when $n_- = 0$: no direction lowers the energy.

17

unstable results rejected

during the production run

0

unstable in the final map

all 2,976,750 nodes re-tested

1

Why the jump test is not enough

On $\varphi_1 = -\varphi_2$ the rod is symmetric ($f_y = 0$); the bent-up branch loses stability at a symmetry-breaking bifurcation, not a fold. Newton follows it smoothly, with no jump in q.

2

Inertia of a bordered matrix

A is the tridiagonal P1 form, C the two constraint vectors (Haynsworth). One LDL^T sweep: O(N), about 0.2 ms against an 11 ms solve.

3

Every candidate

Warm starts and cold starts alike: an unstable result counts as a failure and the node moves on to the next neighbour.

Deriving the exact first mode used for the cold start

1. $\varphi_1 = \varphi_2 = 0$ is mirror-symmetric, so $f_y = 0$ and the elastica is a pendulum

$$\theta'' + \lambda^2 \sin\theta = 0, \quad \lambda^2 = f_x, \quad f_y = 0$$

2. Multiply by θ' and integrate once; $\theta' = 0$ at the peak angle α

$$\theta'^2 = 4\lambda^2\left(k^2 - \sin^2\tfrac{\theta}{2}\right), \qquad k = \sin\tfrac{\alpha}{2}$$

3. Substituting $\sin(\theta/2) = k \sin\psi$ gives $\psi' = \lambda\sqrt{(1 - m \sin^2\psi)}$, so $\sin\psi = \mathrm{sn}(\lambda s \mid m)$

$$\theta(s) = 2\sin^{-1}[k\,\mathrm{sn}(\lambda s \mid m)], \qquad m = k^2$$

4. $\theta(L) = 0$ and $\int \sin\theta\, ds = 0$ need one full period of sn (4K): four quarter-waves

$$\lambda L = 4K(m) \quad \Rightarrow \quad f_x = \left(\frac{4K(m)}{L}\right)^2$$

5. The end distance fixes m; d/L falls monotonically in m, so a 1-D root-find is safe

$$\frac{d}{L} = \frac{1}{L}\int_0^L \cos\theta\; ds = \frac{2E(m)}{K(m)} - 1$$

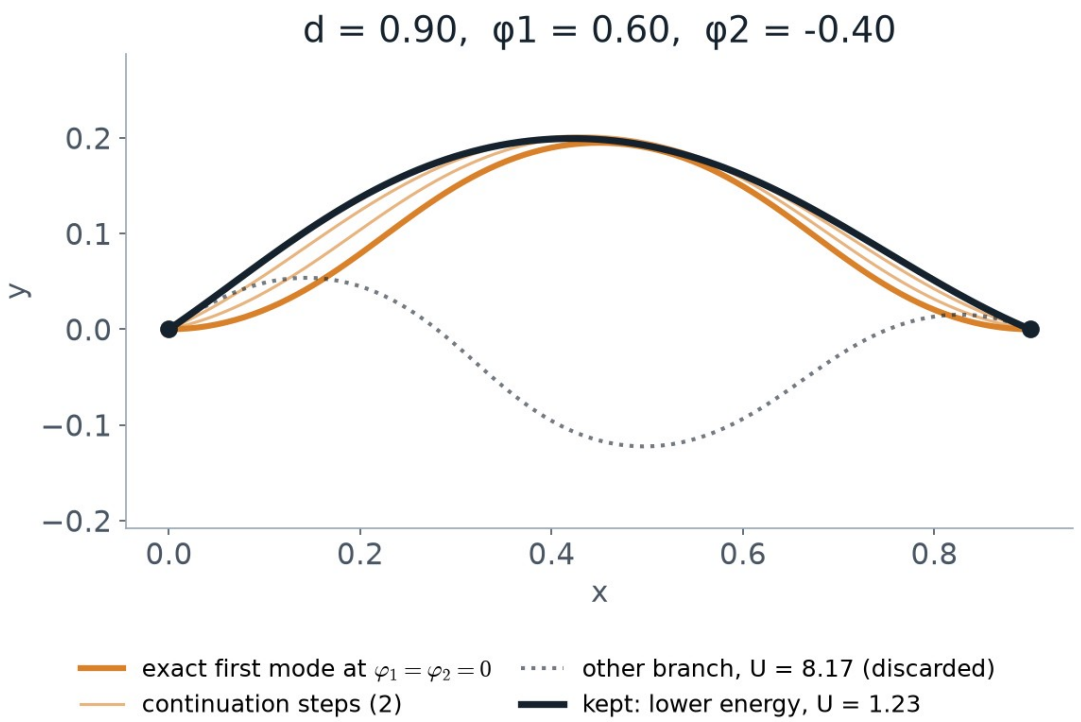
6. Small bow, $m \to 0$: the $A \sin(2\pi s/L)$ shape with $A = 2\sqrt{(1 - d/L)}$, and the Euler load

$$\theta \approx 2\sqrt{m}\,\sin\frac{2\pi s}{L}, \quad m \approx 1 - \frac{d}{L}, \quad f_x \to \frac{4\pi^2}{L^2}$$

$$K(m) = \int_0^{\pi/2} \frac{d\psi}{\sqrt{1 - m\sin^2\psi}}, \qquad E(m) = \int_0^{\pi/2} \sqrt{1 - m\sin^2\psi}\; d\psi$$

Exact first-mode cold start, then continue the clamp angles

$$\theta \approx 2\sqrt{m}\,\sin\frac{2\pi s}{L}, \quad m \approx 1-\frac{d}{L}, \quad f_x \to \frac{4\pi^2}{L^2}$$



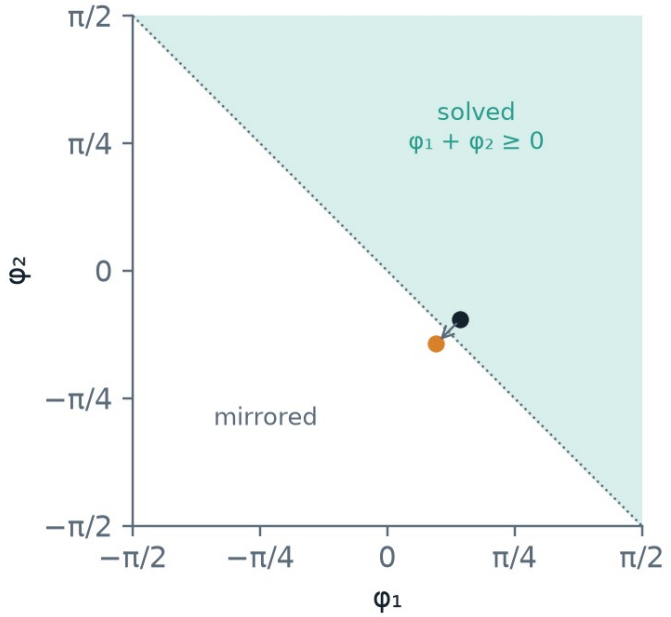
Predictor for each continuation step

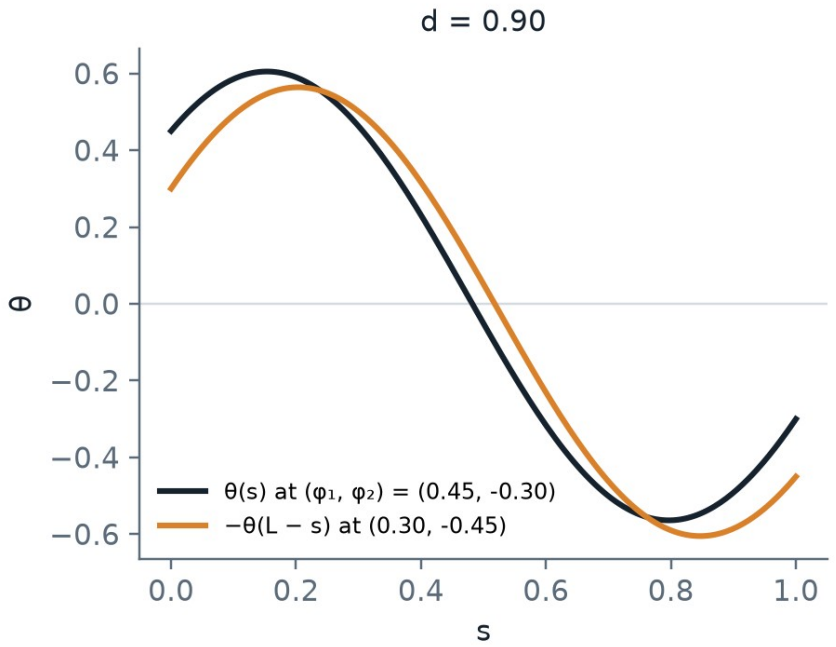
$$\theta_0^{(k+1)}(s) = \theta^{(k)}(s) + \Delta\varphi_1\left(1-\frac{s}{L}\right) + \Delta\varphi_2\,\frac{s}{L}$$

1. Exact at $\varphi_1 = \varphi_2 = 0$ for any d, with $\int \sin\theta\, ds = 0$: Newton converges in 2 iterations
2. Steps ≤ 0.25 rad, ≤ 15 Newton iterations each; a failed step halves, giving up below 0.02 rad (a fold)
3. Both hump signs are continued and the lower energy kept, so the result does not depend on which branch the ramp follows

Mirror symmetry halves the number of solves

$$(d, \varphi_1, \varphi_2) \mapsto (d, -\varphi_2, -\varphi_1): \quad \tilde{\theta}(s) = -\theta(L-s), \quad \tilde{f}_x = f_x, \quad \tilde{f}_y = -f_y, \quad \tilde{M}_1 = M_2, \quad \tilde{p} = p$$





1. Reflecting the rod about mid-span maps (φ_1, φ_2) to $(-\varphi_2, -\varphi_1)$
2. Only $\varphi_1 + \varphi_2 \geq 0$ is solved; points on the anti-diagonal are solved once
3. f_x, U and p carry over, f_y flips sign, and the left and right moments swap

Final run

$d \in [0.70, 0.99]$, $\varphi_1, \varphi_2 \in [-\pi/2, \pi/2]$, $\Delta = 0.01$: 2.98 M equilibria and their level sets

2.98 M equilibria, every one checked

2,976,750

equilibria

30 × 315 × 315 grid; 1.49 M Newton solves, the mirror fills the rest

≈ 4 h

on one core

about 11 ms per solve, from one cold start (plus one more near d = 1)

0.27 %

max error vs. BVP

300 random nodes re-solved with scipy solve_bvp; median 0.014 %

0

unstable equilibria

stability test on all 2,976,750 nodes

Exact checks pass

Mirror pairs agree to 3×10^{-8}. At $\varphi_1 = \varphi_2 = 0$ every layer matches the exact first mode: f_x within 1.3×10^{-4}.

Branch share

Bent up on 70.6 % of the layer at d = 0.70, falling almost linearly to 50.5 % at d = 0.99. Snap surface area 2.64 (30,081 snap edges).

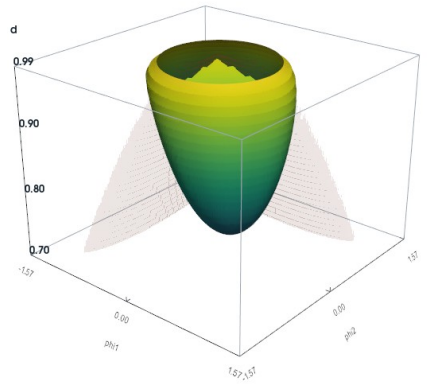
Ranges

f_x from −13,785 (tension, d = 0.99, $|\varphi| = \pi/2$) to +93; $|f_y| \le 340$; $|M| \le 168$; U from 0.10 to 142.

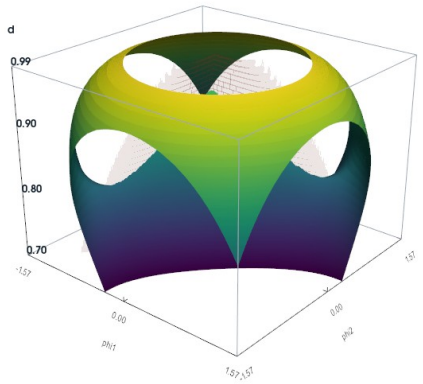
ParaView: data/production/post/elastica_solution_space.vtu (every node, all fields).

Constant-energy surfaces: shells around the straight rod

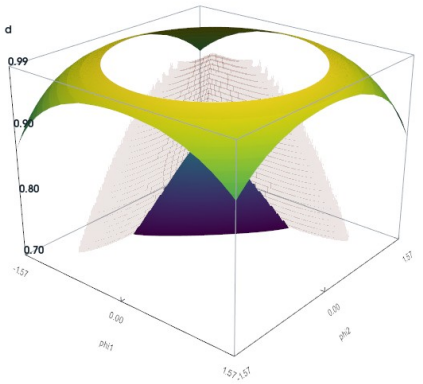
U = 2



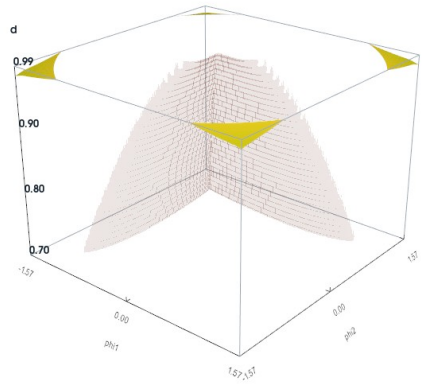
U = 8



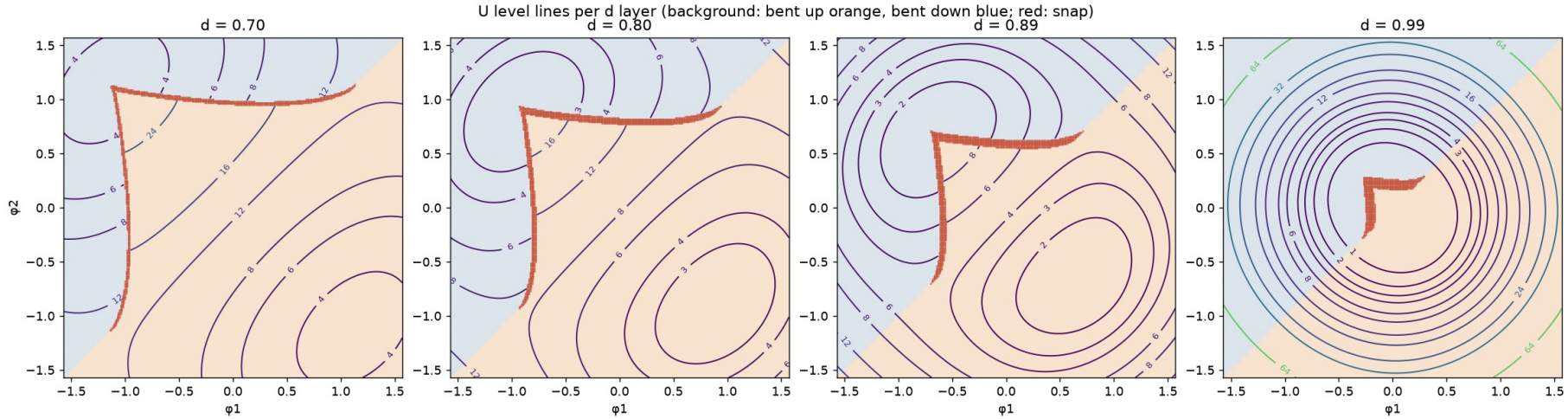
U = 16



U = 64



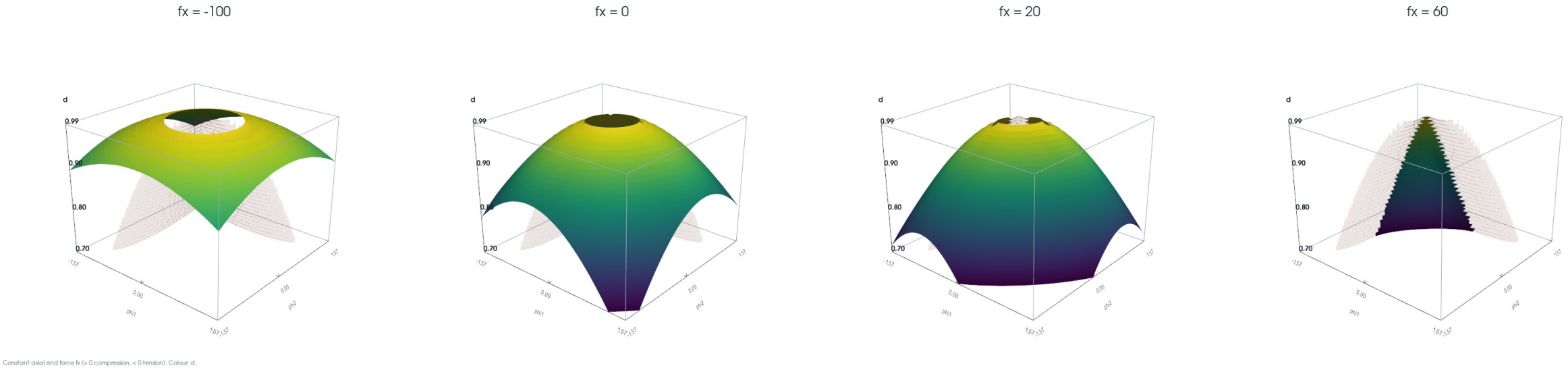
Constant bending energy U = 1/2 ∫ theta'^2 ds (B = L = 1). Colour: d. Faint red: snap part of the p = 0 surface; level sets stop at it.

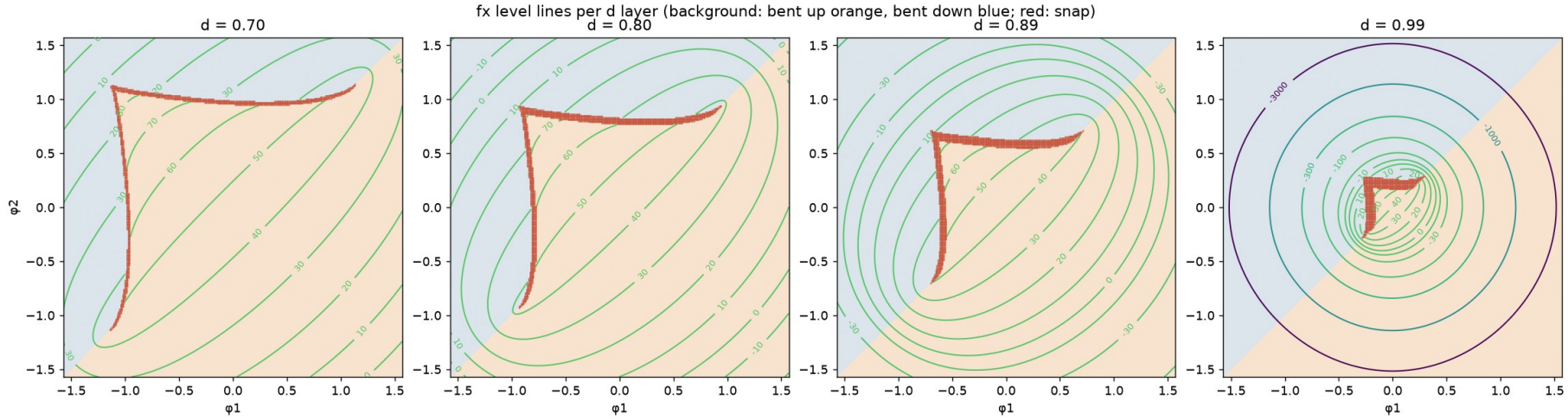


$U = \frac{1}{2}\int \theta'^2\, ds$

From 0.10 to 142. Lowest for a nearly straight rod ($d \to 1$, $\varphi \to 0$): U = 2 is a bowl around that corner and higher levels are wider domes. Each ends at the snap (faint red), where U jumps.

Constant axial force f_x: bounded in compression

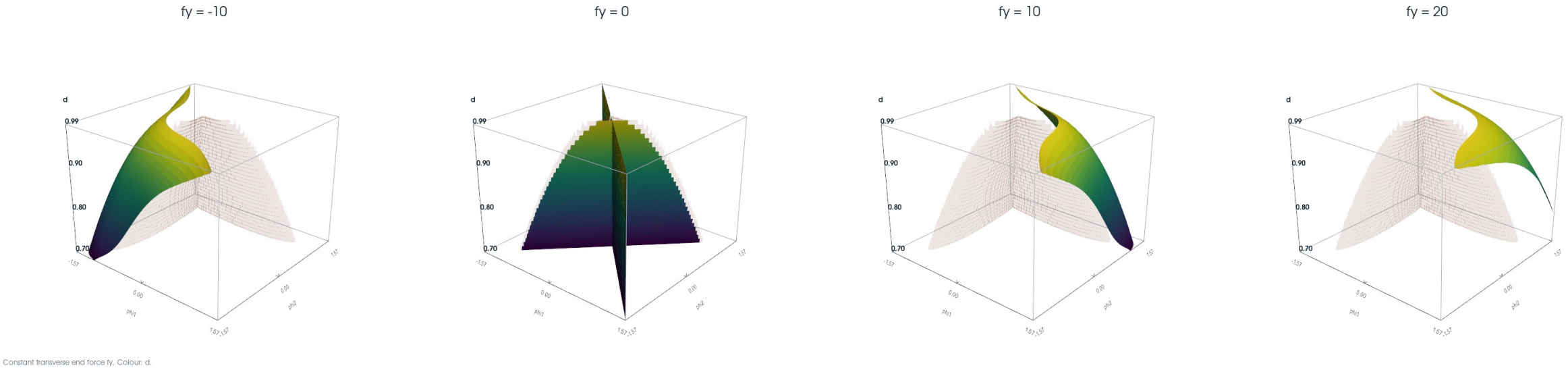




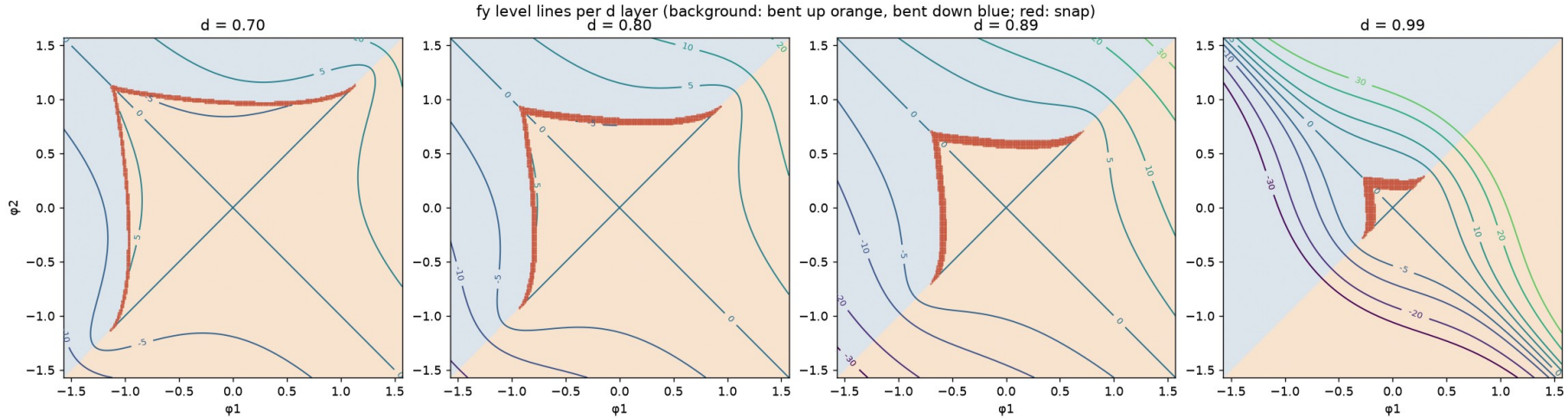
$f_x > 0$: compression

From −13,785 (tension, d = 0.99, $|\varphi| = \pi/2$) to +93. Compressive levels exist only inside the bent-up region: $f_x = 60$ is a sliver and nothing reaches 100.

Constant transverse force f_y: mirror-image pairs



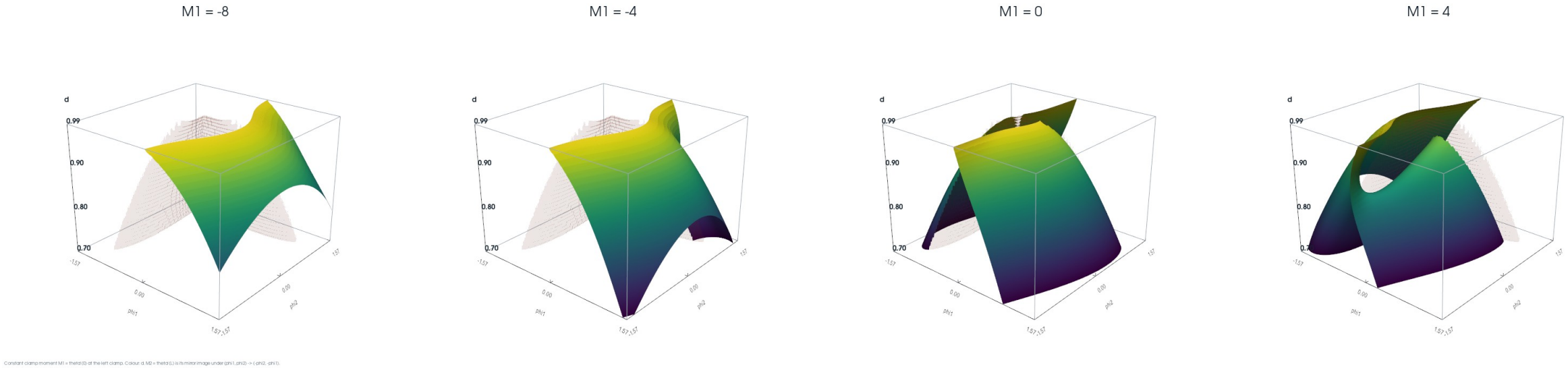
Constant transverse end force fy. Colour: d.

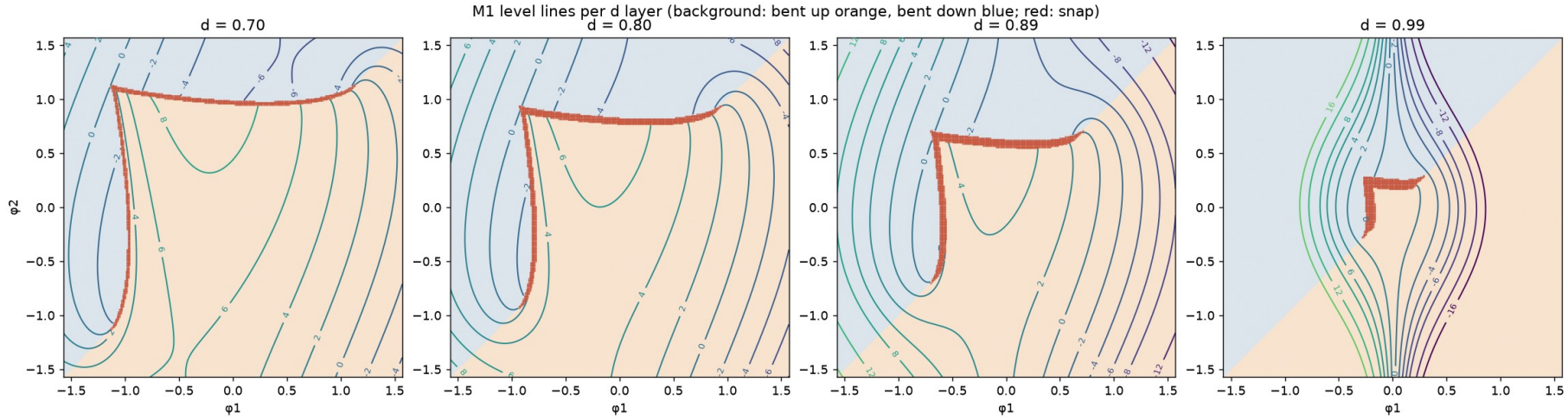


Odd under the mirror

$|f_y| \le 340$. $(\varphi_1, \varphi_2) \to (-\varphi_2, -\varphi_1)$ flips f_y, so $f_y = 0$ holds on the whole plane $\varphi_1 = -\varphi_2$ and ± levels come in pairs of equal area.

Constant clamp moment M_1; M_2 is its mirror image





$M_1 = \theta'(0)$

Within ±168. M_2 at (φ_1, φ_2) is M_1 at $(-\varphi_2, -\varphi_1)$, so its surfaces are mirror images (both in the VTU). Small $|M_1|$ levels split into two sheets.

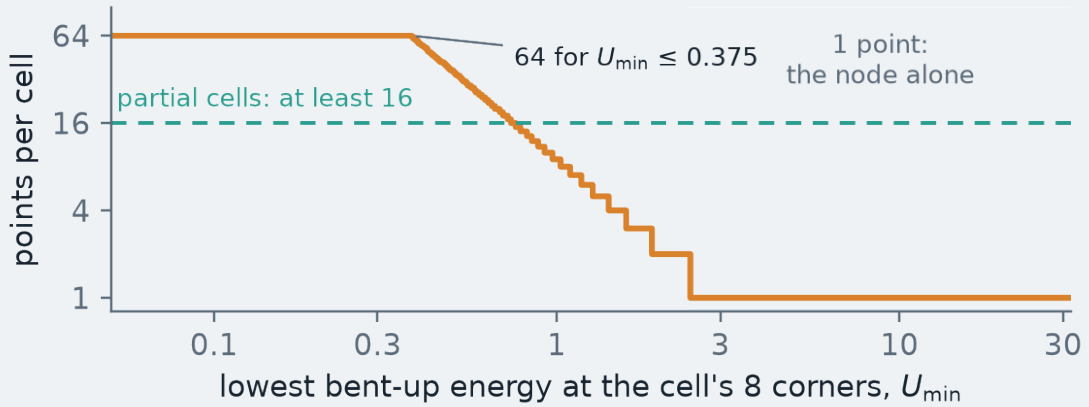
Refined sampling

Training data: both branches at every node, and more solves where the energy is low: 7.6 M states

More points per cell where the energy is low

Points per grid cell

$$n = \mathrm{clip}\left(\left(\frac{u_{\mathrm{ref}}}{U_{\min}}\right)^2,\ 1,\ 64\right)$$



1.11 M

new points solved

998 k states, 112 k past a fold; 23 min on 14 workers

7.6 M

states in the dataset

28.9 % with U < 1, against 2.4 % in the production run

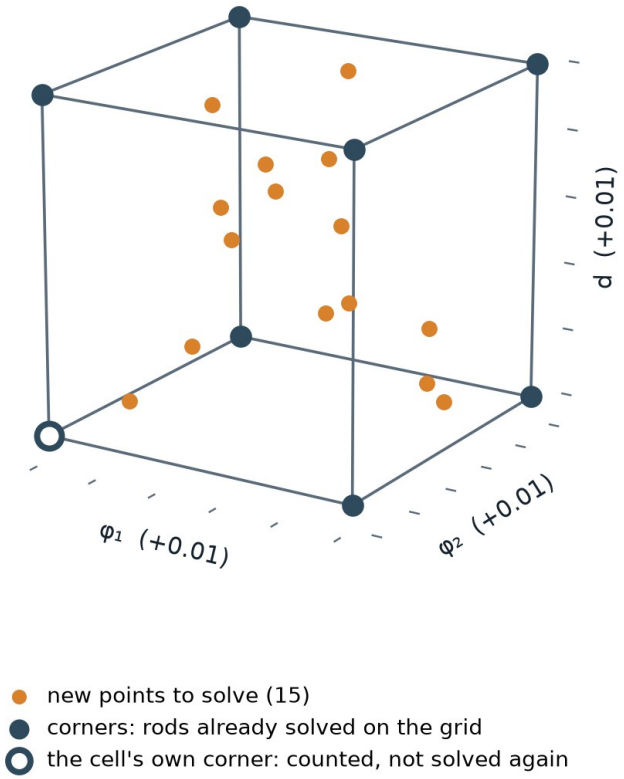
1. **Density follows the energy**
   U_{min} is the lowest bent-up energy at the cell's 8 corners and u_{ref} = 3: one point (the node) above U_{min} = 2.45, all 64 below 0.375.

2. **Partial cells: at least 16**
   Cells where the bent-up branch holds at some corners only: its fold edges, and along $\varphi_1 = \varphi_2$, where p changes sign smoothly.

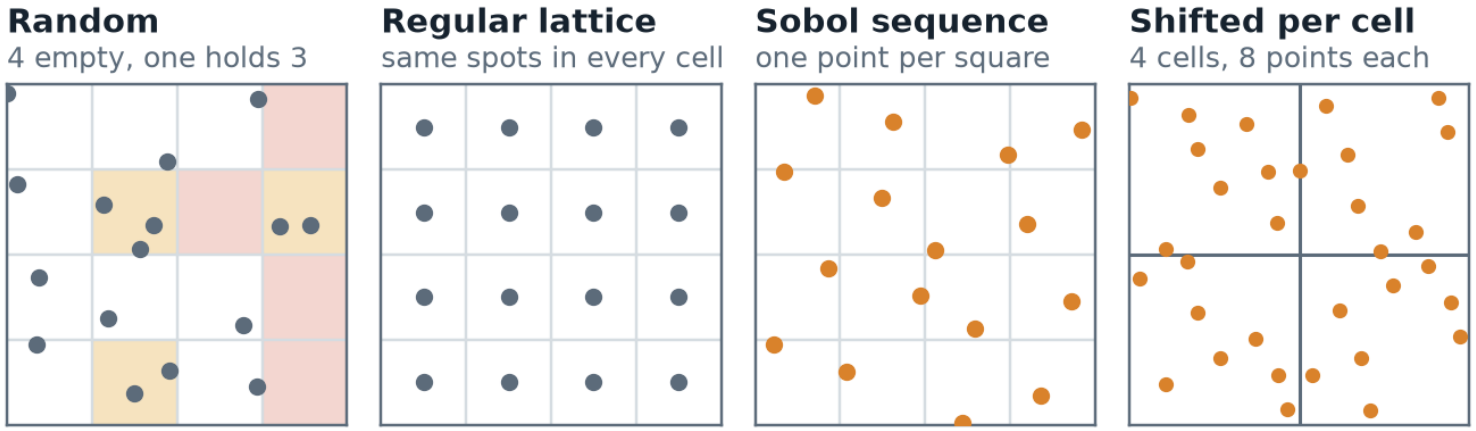
3. **Spread through the cell**
   The node, then a scrambled Sobol set with a random shift per cell. Only the bent-up branch on $\varphi_1 + \varphi_2 \ge 0$ is solved; mirror and reflection give the rest.

4. **Production tests, and negatives**
   Warm start from the corners, nearest first, at most 3 tries: no jump in q, stable. A failure in a partial cell is kept as a negative example (no state).

Inside a cell: new points spread evenly



Example: n = 16, so the cell's own corner plus 15 new points, anywhere inside, d included: the data also covers values between the 30 d layers.



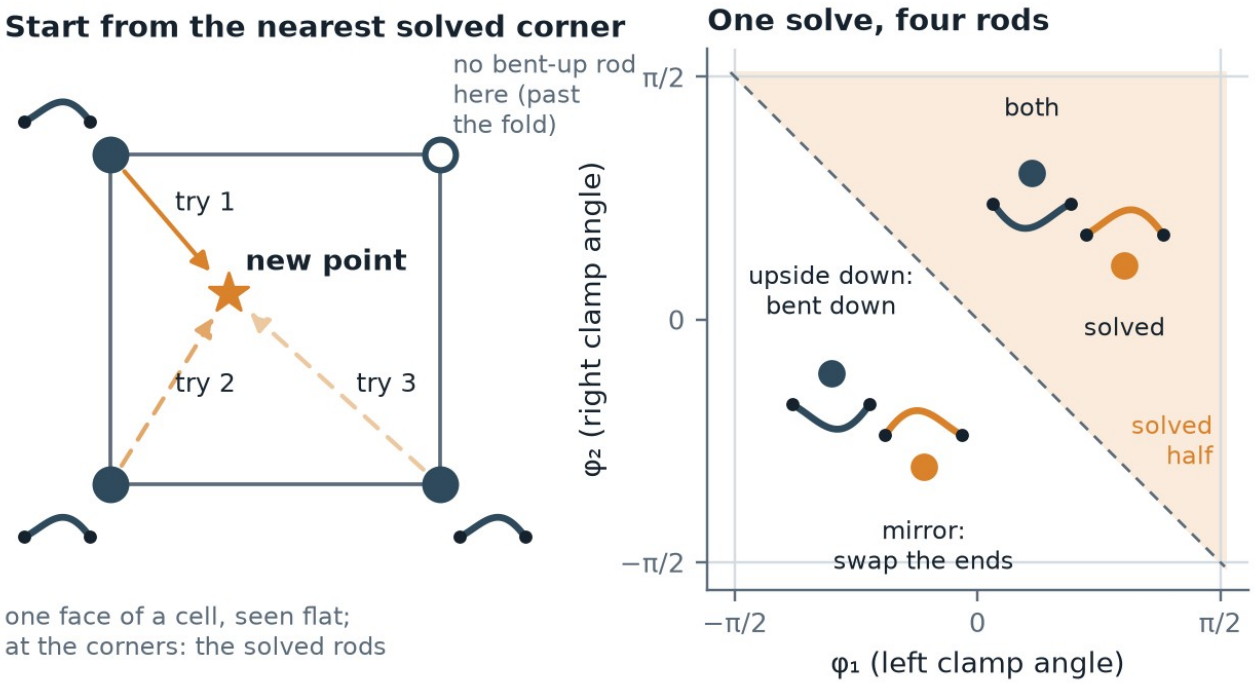
16 points in one cell, seen flat (φ_1 across, φ_2 up). Red: empty square; yellow: crowded square

1 **What a cell is**
The box between 8 neighbouring grid points, 0.01 wide in φ_1, φ_2 and d. Its corners are already solved.

2 **Spread evenly: Sobol**
A recipe that gives every part of the box its share of points: no clumps, no holes.

3 **A new shift in each cell**
Each cell slides its pattern by its own random amount, so neighbouring cells never share one pattern.

Solving each new point from its neighbours



998 k

rods accepted

from 1.11 M new points

112 k

“no rod here”

kept: past a fold

9

dropped

3 more: solved afresh

1 **Start from a nearby answer**

Each rod is found by Newton's method, which improves a guess step by step. A solved corner rod, at most 0.01 away, is a very good guess.

2 **Nearest corner first, up to 3 tries**

Only corners holding the bent-up rod are used. The answer is kept if it converged, did not snap to the other shape, and is stable.

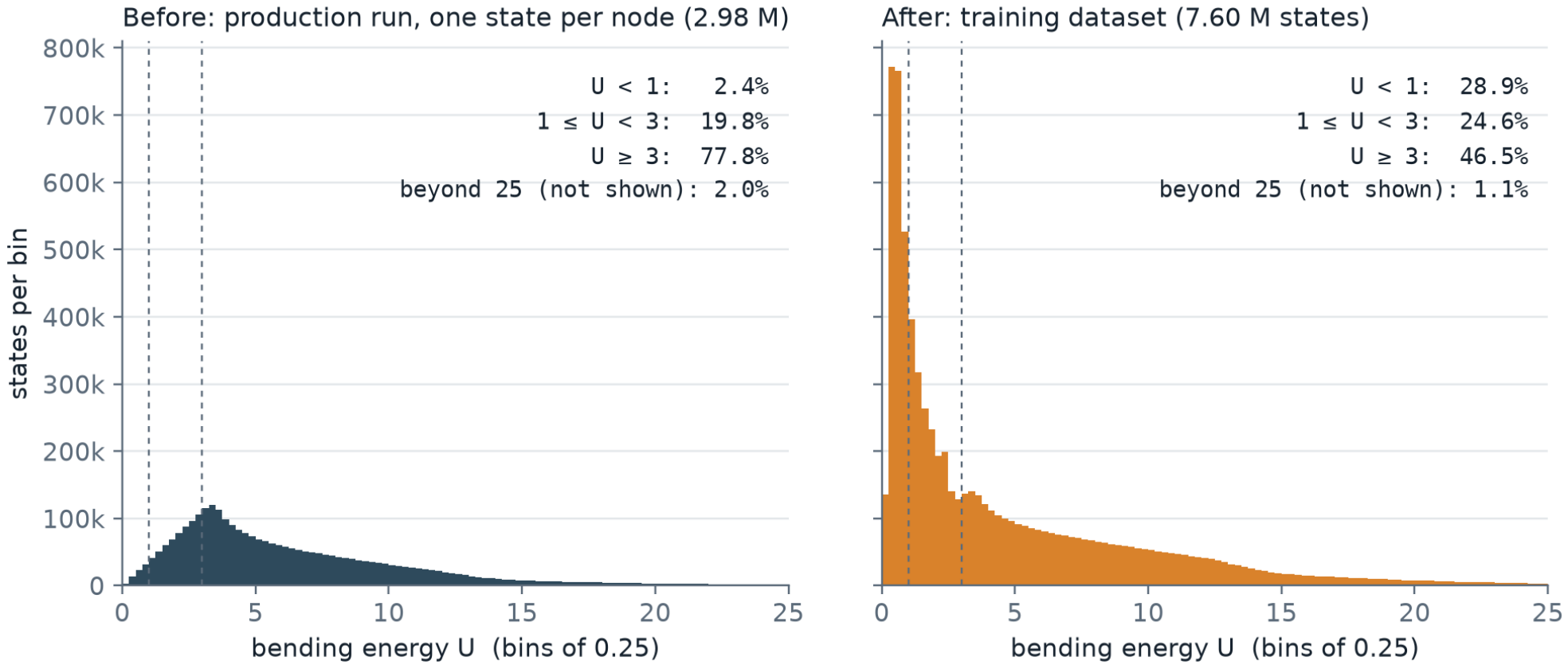
3 **If every try fails**

Near a fold the bent-up rod may not exist at that point: it is kept as a “no rod here” example, which shows where the shape ends. Elsewhere: a fresh solve, or the point is dropped.

4 **One solve gives four rods**

A rod seen from behind (ends swapped) or turned upside down is still a valid rod, so only a quarter of the points are solved.

Refinement moves the data to low energy

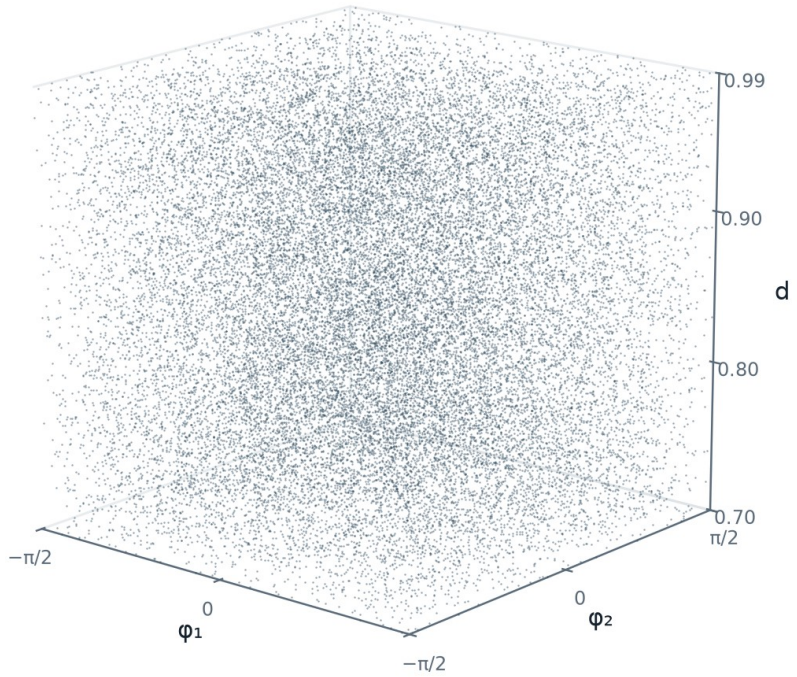


Same bins and axes. Before: one state per grid node, 2.4 % of them with U < 1. After: both branches at every node plus 4.0 M refined states (998 k solves and their symmetry images); 28.9 % with U < 1 and 24.6 % with 1 ≤ U < 3. The grid's high-energy states are all kept.

In 3D: the new points gather where the energy is low

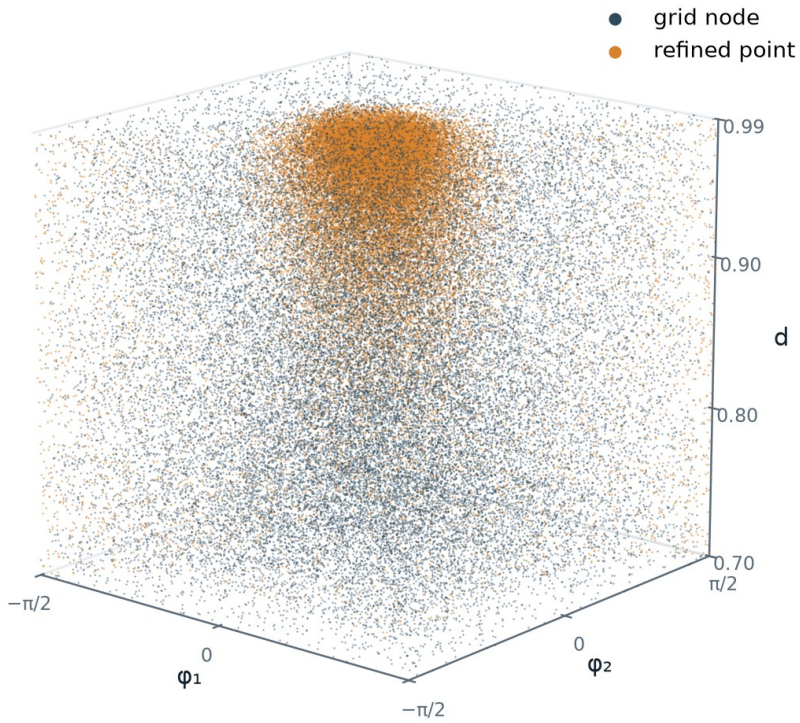
Before: 2.98 M grid states

one per grid node; 1 % drawn (29,897 dots)



After: 7.60 M states

both branches + refined points; 1 % drawn (76,088 dots)

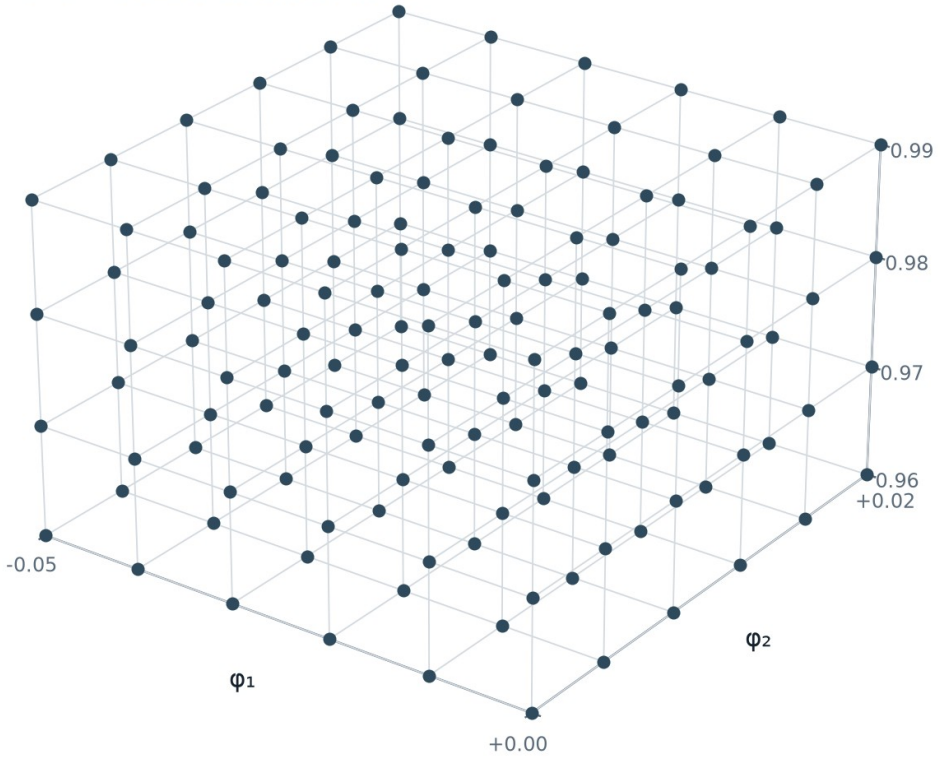


Each side draws the same random 1 % of its points, so dot density is data density. Before: one state per grid node, evenly spread. After: refined points (orange) fill the low-energy region at high d near the centre; grid nodes (slate) still cover the rest. Orange away from the centre comes from the minimum of 16 points per partial cell: fold edges, and the diagonal $\varphi_1 = \varphi_2$.

Close up: more points inside the cells where U is low

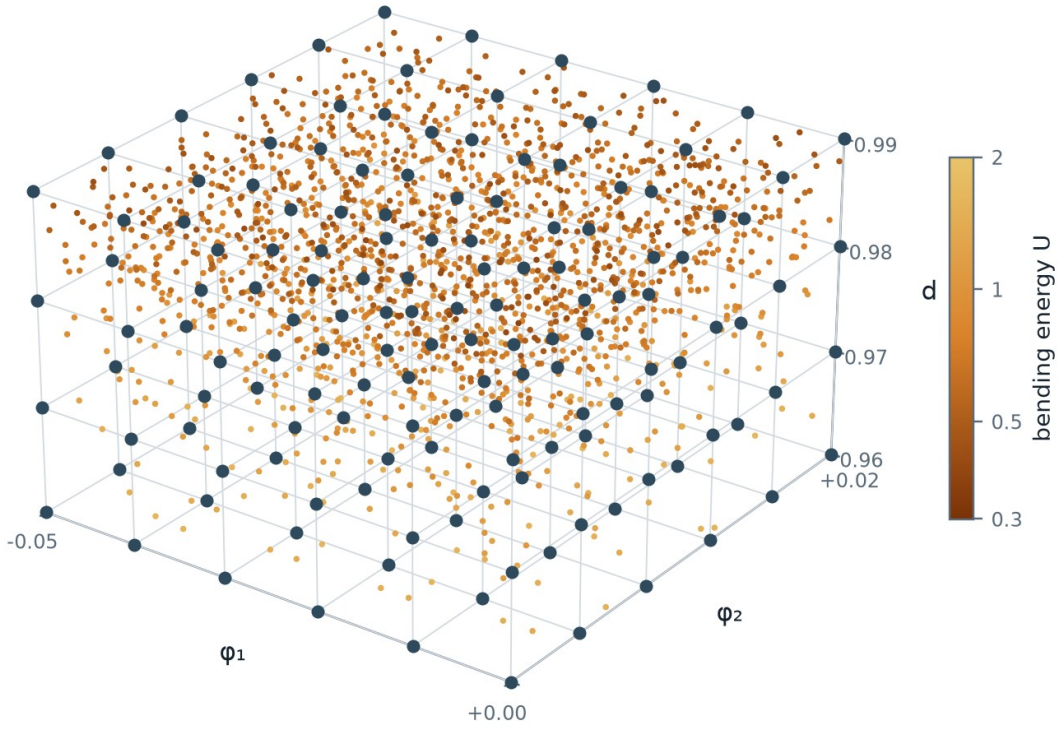
Before: 144 grid nodes

5 × 5 × 3 cells, a rod at every corner



After: 1,926 bent-up states

the same corners, plus points inside each cell

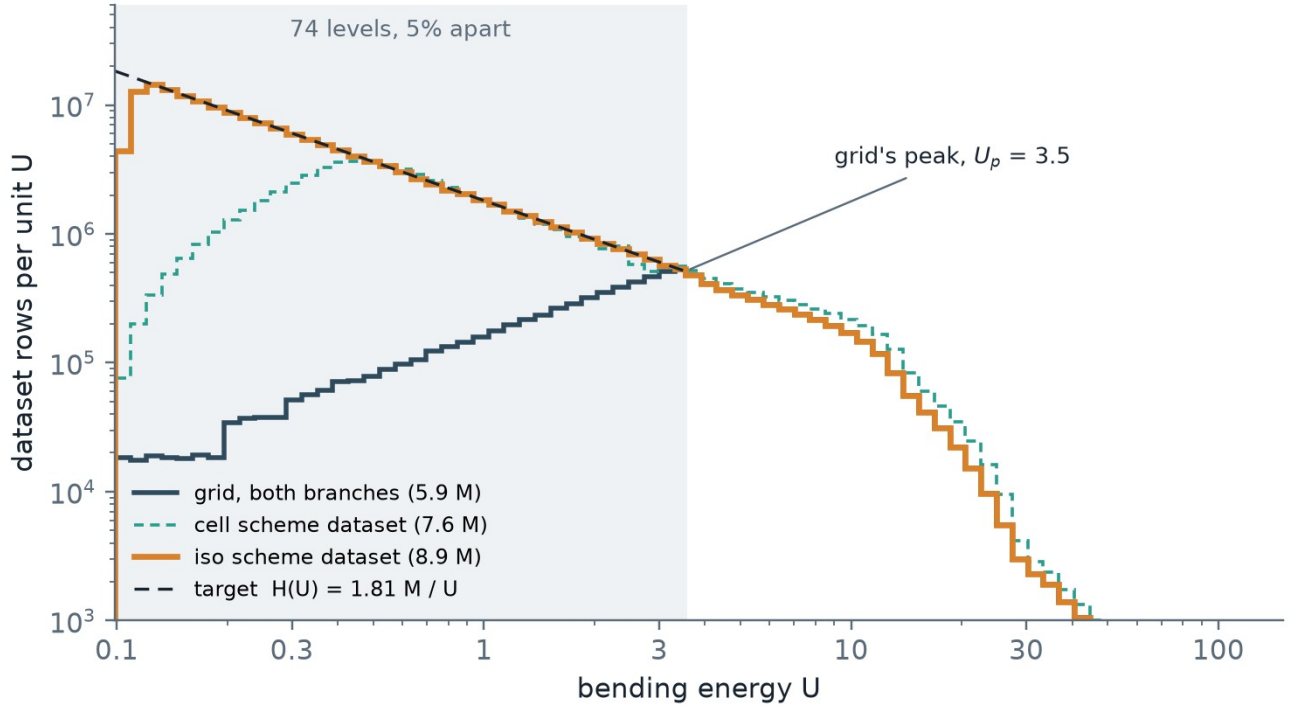


Every bent-up point in 75 cells at d = 0.96–0.99, $\varphi_1 \in [-0.05, 0]$, $\varphi_2 \in [-0.03, 0.02]$. The grid nodes stay; each cell adds its own points inside. Towards the top U falls (0.36 at the lowest) and the cells hold up to 64 points; at the bottom, where U reaches 1.8, as few as 5.

Iso-energy sampling

A second scheme: points on level sets of U, as many per level as a 1/U target asks: 8.9 M states, 45 % with U < 1

Aim the whole histogram at a target



74
levels
U = 0.10 to 3.5

1.33 M
points solved
none dropped

1.81 M / U
rows per unit U
measured, 0.12 ≤ U ≤ 3.5

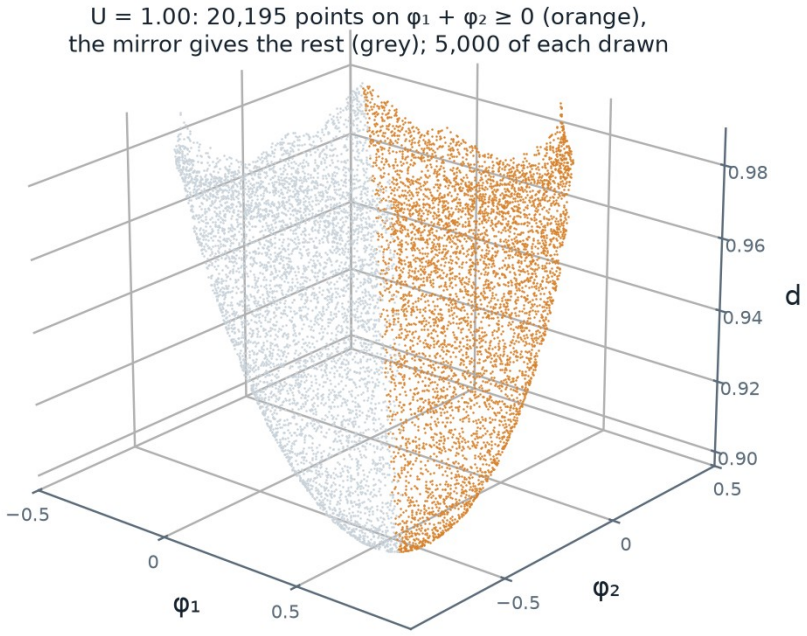
1 **A target for the dataset**
Rows per unit energy H(U) = C / U, from the lowest energy, 0.10, up to U_p ≈ 3.5, where the grid's own histogram peaks. Above U_p the grid alone.

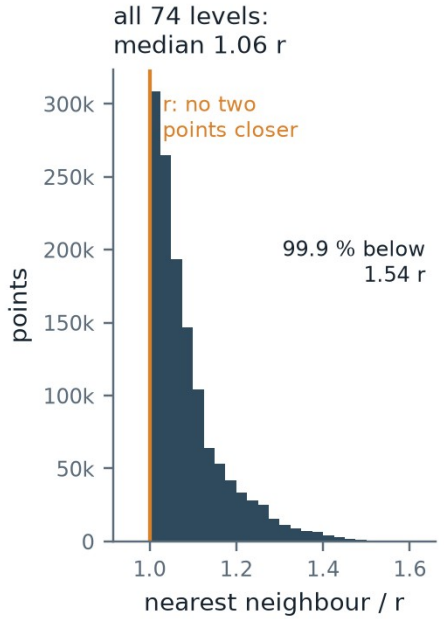
2 **Why 1 / U**
Every decade of U gets the same data: what a fit of log U, or of U to a fixed relative error, needs. 1 / U^2 would take 15.5 M solves, 11 times as many.

3 **Levels 5 % apart**
74 levels U_k from 0.10 to 3.5. Each gets the target's rows in its interval, less the grid's rows already there, divided by four: one solve gives four rods.

4 **Planned before any solve**
A dry run places every point and predicts the histogram: 2.5 min for all 74 levels.

Points spread evenly over each level surface





1.06 r

median spacing

nearest neighbour, all levels

1.54 r

99.9 % closer

no wide gaps

2.6 k

on the lowest level

target 22 k: a tiny cap at d = 0.99

1

The level surface

Marching tetrahedra over the 6 tetrahedra of every cell, only those with the bent-up rod at all four corners and no jump in q: the surface ends at the fold.

2

Candidates

The surface's own vertices, one per crossed grid edge, then a lattice inside every triangle at half the spacing.

3

Poisson-disk selection

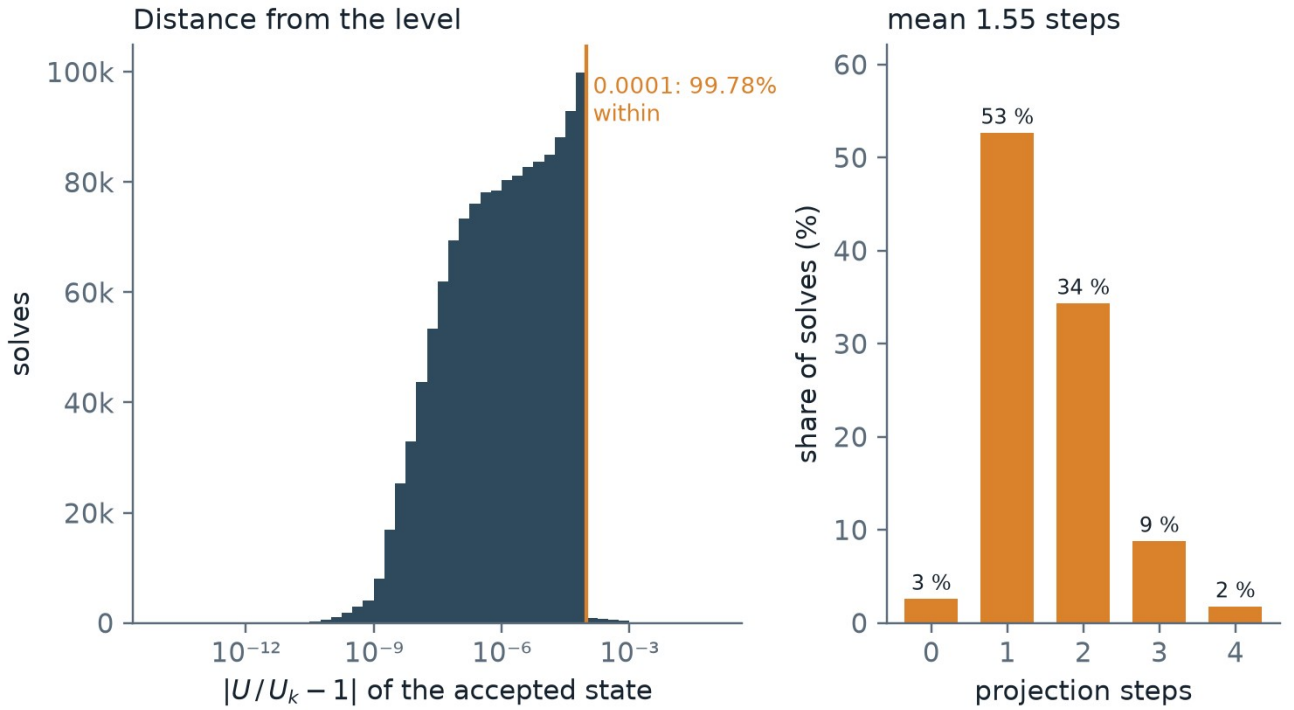
Vertices first, then lattice points, in random order: a point is kept unless a kept point lies within r. r is tuned per level to within 3 % of its target.

4

Half the surface

Only $\varphi_1 + \varphi_2 \geq 0$ is placed; the mirror and the reflection give the other three rods. r never drops below 0.05 grid steps.

Solve, then slide each point onto its level



99.78 %

within 10^{-4}

the rest: valid rods, 2.8×10^{-4} off

1.55

projection steps

mean per solve

28 min

on 14 workers

1.33 M solves

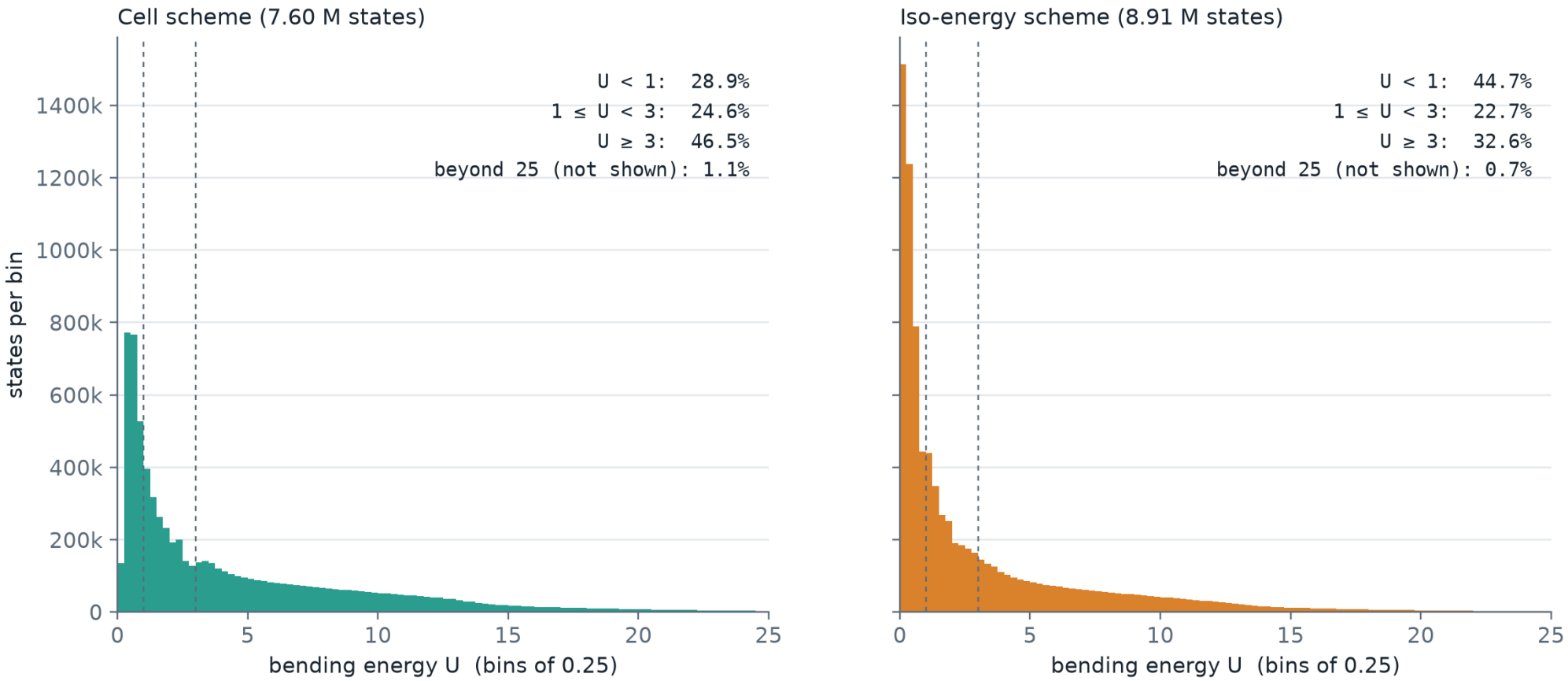
1. **Warm start from the corners**
   As in the cell scheme: corners holding the bent-up rod, nearest first, at most 3 tries; no jump in q, and stable.

2. **It lands near the level**
   The surface came from linear interpolation, so the solve misses U_k slightly; near d = 0.99 by a few %.

3. **The gradient comes free**
   Envelope theorem: $\nabla U = (-fx, -M_1, M_2)$ in $(d, \varphi_1, \varphi_2)$, which the solve already gives. Step by $-(U - U_k)\,\nabla U / |\nabla U|^2$ and re-solve from the rod just found.

4. **Up to 4 steps**
   Until $|U / U_k - 1| < 10^{-4}$. A step that would leave the box slides along its face.

Iso-energy sampling puts 45 % of the data below U = 1

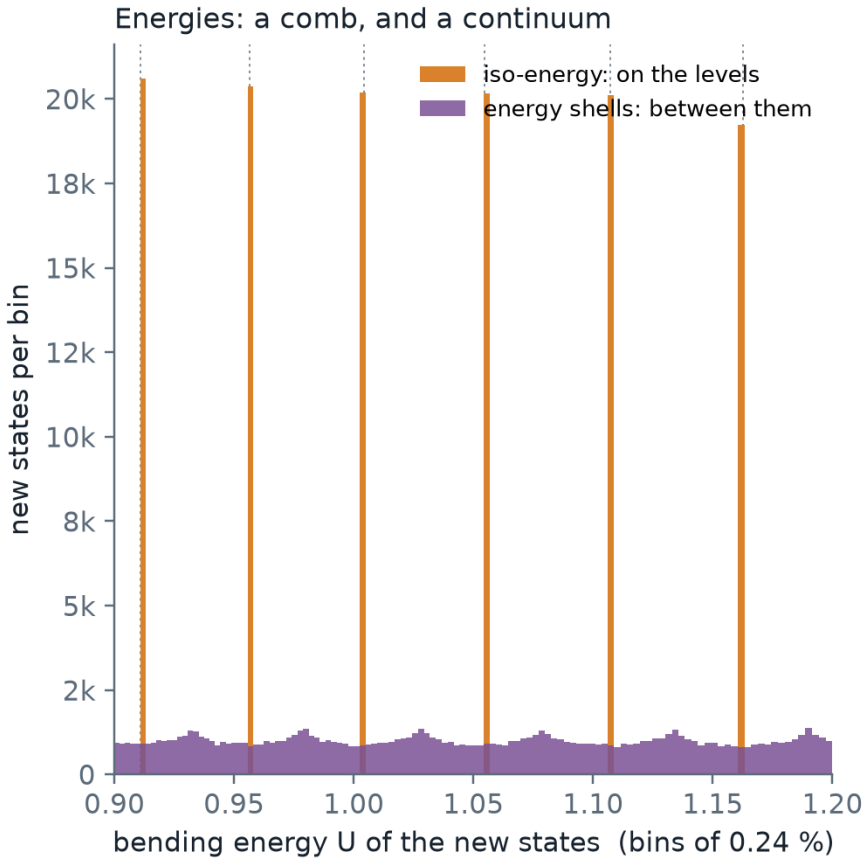


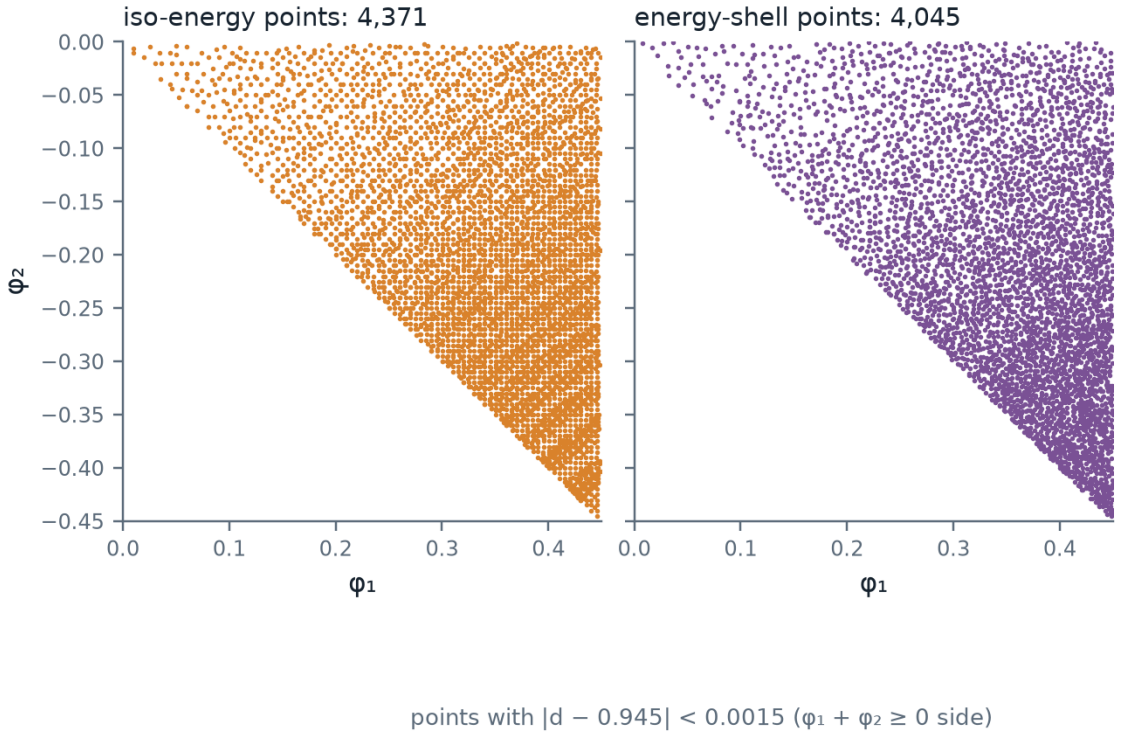
Same bins and axes as the cell scheme. The iso scheme adds 5.3 M states (1.33 M solves and their images), 74 % of them with U < 1; the grid's states are all kept. The lowest four levels, within 20 % of the lowest energy, are tiny caps on the d = 0.99 face and hold 2.6 k–21 k points instead of 22 k.

Energy-shell sampling

A third scheme: the same target, with the points spread through the shells between the levels instead of on them: 9.0 M states, 46 % with U < 1

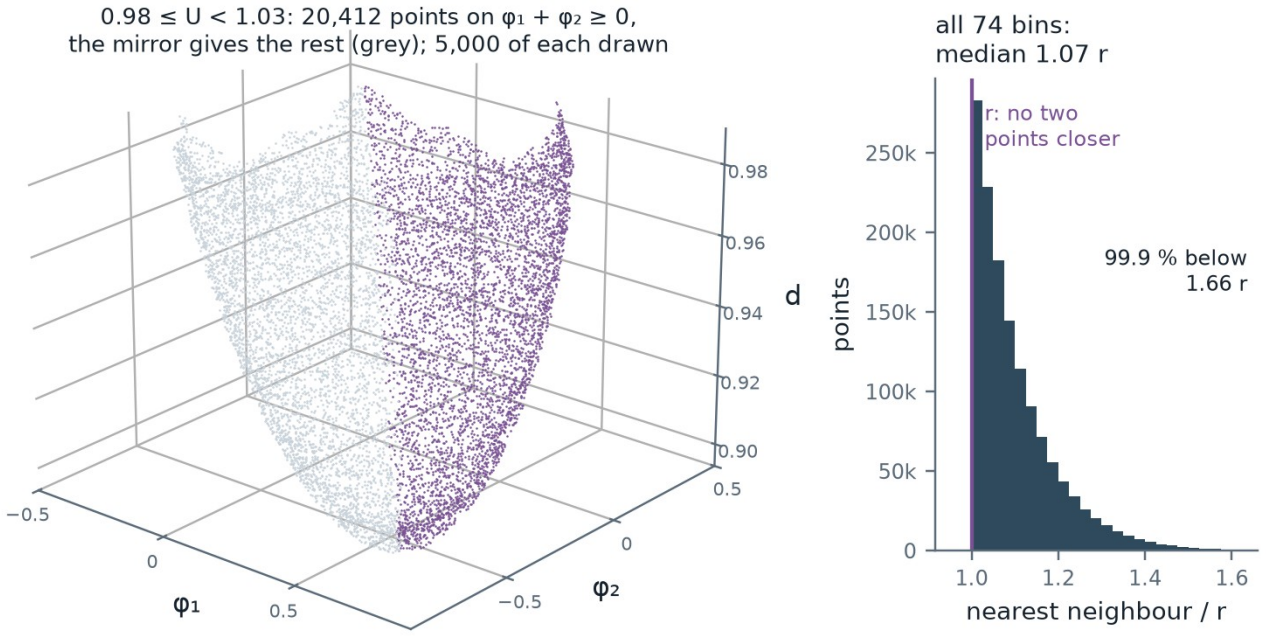
Points on the levels leave the energies between them empty





Left: energies of the new states from U = 0.9 to 1.2, on 0.24 % bins. The iso scheme's sit on its levels (dotted), 5 % apart: all of them within 0.5 % of a level. The shells' fill the range: 18 % within 0.5 % of a level, about what evenly spread energies give. Right: in space the two look alike at this scale; a slab of d a third of a grid step thick holds 4.4 k and 4.0 k points. The gap is in the energies.

Fill the shell between two levels



1.36 M

points solved

planned in 69 s, none dropped

99.64 %

within 10^{-4} of plan

99.95 % in their bin

29 min

on 14 workers

1.62 projection steps on average

1 **Same target, levels as bin edges**

H(U) = 1.81 M / U up to $U_p \approx 3.5$, in 74 bins 5 % wide; each gets the target's rows less the grid's, divided by four.

2 **The shell**

Where the energy interpolant over the jump-free bent-up tetrahedra lies in the bin: it ends at the fold and at the box.

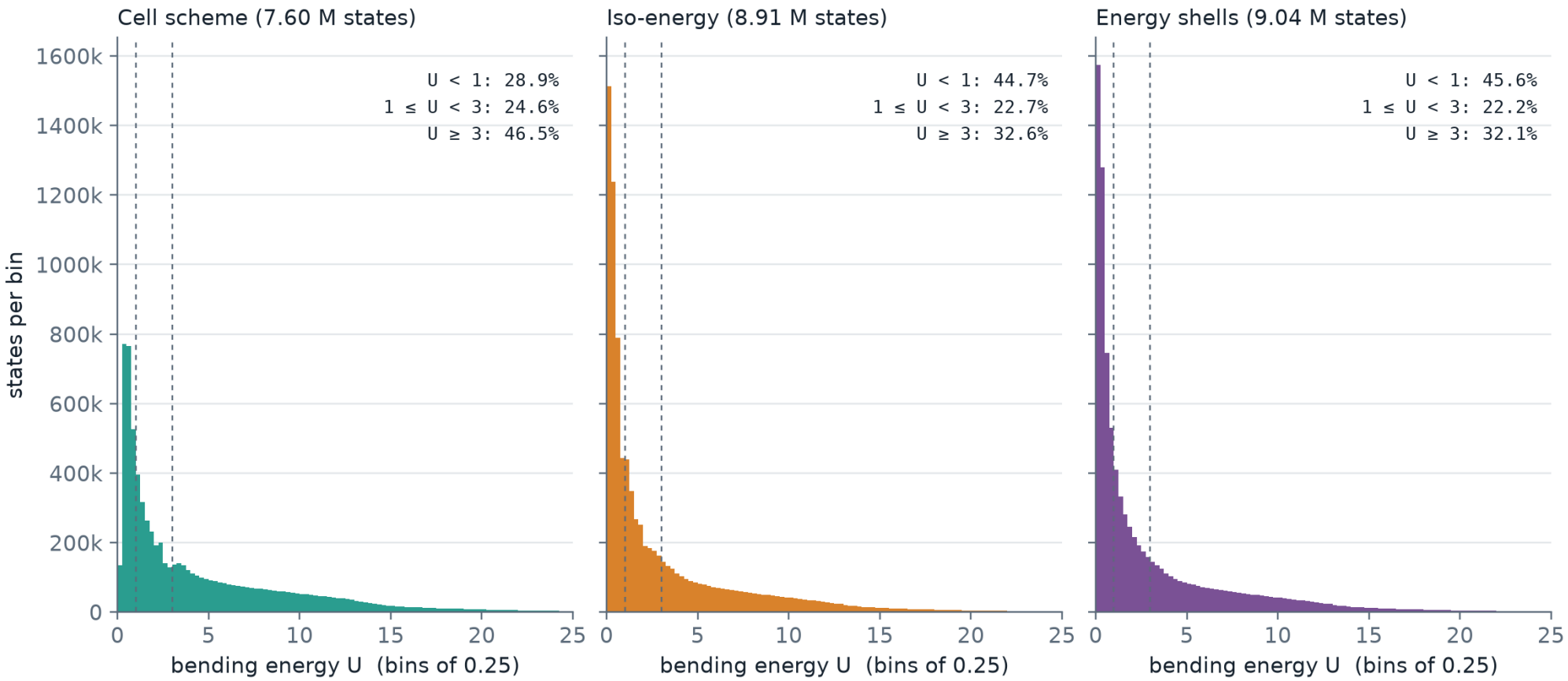
3 **Uniform candidates, Poisson disk in 3D**

The same number of random points in every tetrahedron the bin crosses, kept inside the shell, 12 per point wanted; a greedy 3D Poisson-disk pass keeps the target, r tuned per bin.

4 **Solve, then project onto the plan**

Warm start from the corners, then along $\nabla U = (-fx, -M_1, M_2)$ to the point's planned energy, within 10^{-4}. Stopping at the bin edge instead piles states up there.

Three schemes on the same bins



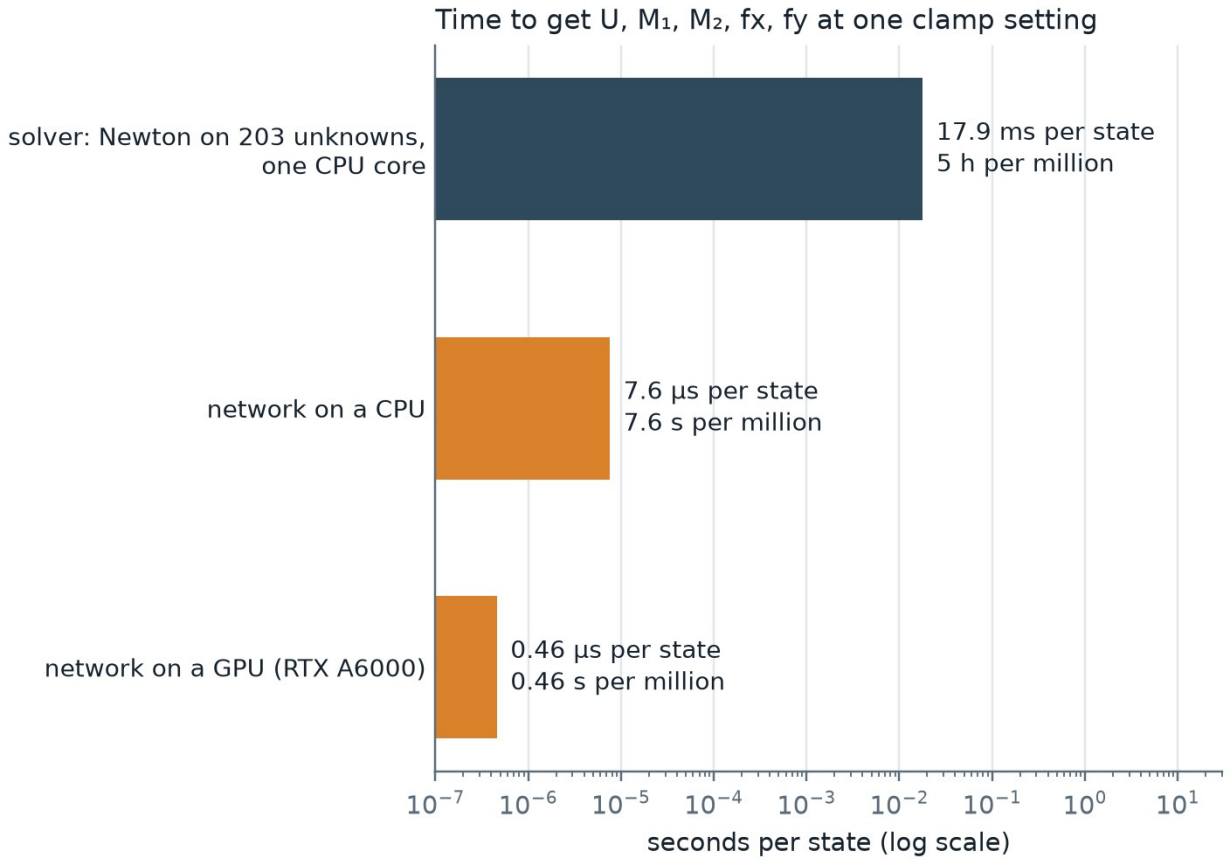
The shells add 5.4 M states (1.36 M solves and their images) and match the iso scheme bin for bin at this scale: 46 % of the data below U = 1, against 29 % per cell. In the smallest bins, caps at d = 0.99, a 3D shell holds more points than a surface: 5.4 k, 11 k and 17 k in the lowest three, against 2.6 k, 8.4 k and 15 k; the target is 22 k.

A neural-network surrogate

A fast formula, fitted to 9 million solver results, that gives U, M_1, M_2, fx, fy and which shapes exist at any clamp setting; and how far it can be trusted

Why replace the solver with a network?

A solve takes 18 ms and must start from a solved neighbour; the network answers any clamp setting in microseconds, with exact derivatives.



1 **The solver**

Newton iterations on 203 unknowns at every point, each started from an already-solved neighbour: 17.9 ms of one CPU core, 5 hours per million states.

2 **The surrogate**

One formula, evaluated at any (d, φ_1, φ_2) and either shape, with no neighbour needed: 0.46 µs per state on a GPU, 7.6 µs on a CPU.

3 **What that is for**

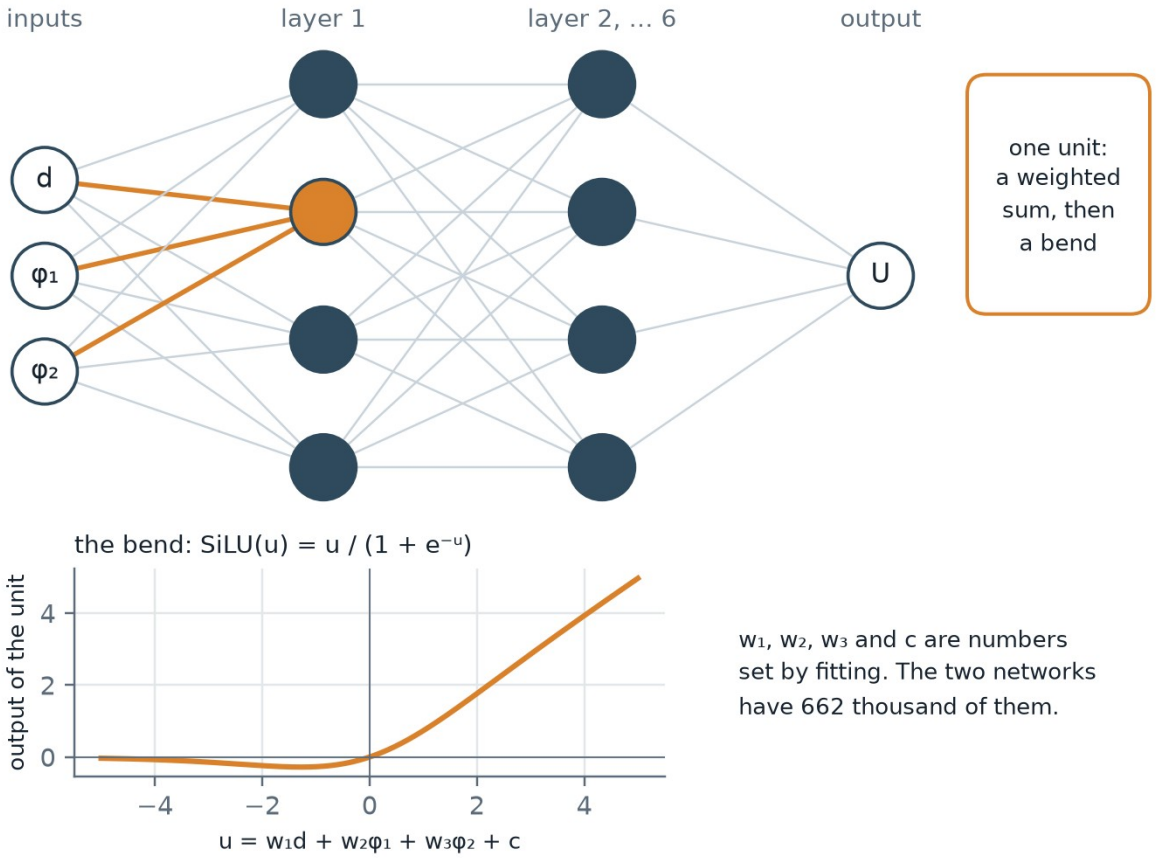
Sweeps over millions of clamp settings, and optimisation or control loops that need many evaluations and their derivatives.

4 **The price**

It is a fit, so it carries an error. The slides that follow measure that error, and where it is largest.

What is a neural network?

A flexible formula built from many simple units. Fitting sets its 662 thousand numbers so that it reproduces the solver’s results.



w1, w2, w3 and c are numbers set by fitting. The two networks have 662 thousand of them.

1 **One unit**

Takes a weighted sum of its inputs, adds a constant, and bends the result with a smooth curve (bottom left).

2 **Layers of units**

Each layer feeds the next: 6 layers of 256 units. Stacking simple bends lets the formula follow curved, steep surfaces such as U(d, φ_1, φ_2).

3 **Fitting**

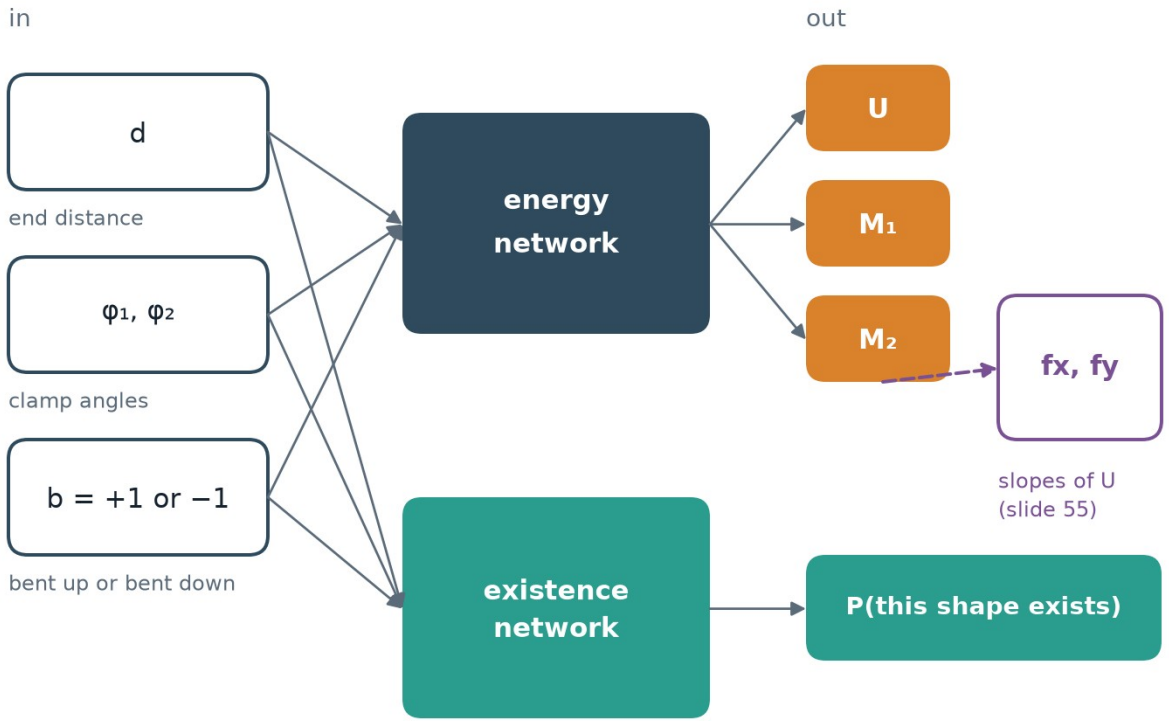
Start from random weights; again and again, compare with the solver on a batch of states and nudge every weight to reduce the mismatch (gradient descent).

4 **Why a smooth bend**

The formula can then be differentiated exactly everywhere, which the forces need (slide 55).

What goes in, and what comes out?

In: the clamp setting and the shape, bent up or bent down. Out: U, M_1, M_2 and whether that shape exists. The forces follow from U.



1 **The shape is an input**
Where both shapes exist, one clamp setting has two answers: b = +1 or −1 says which one is wanted.

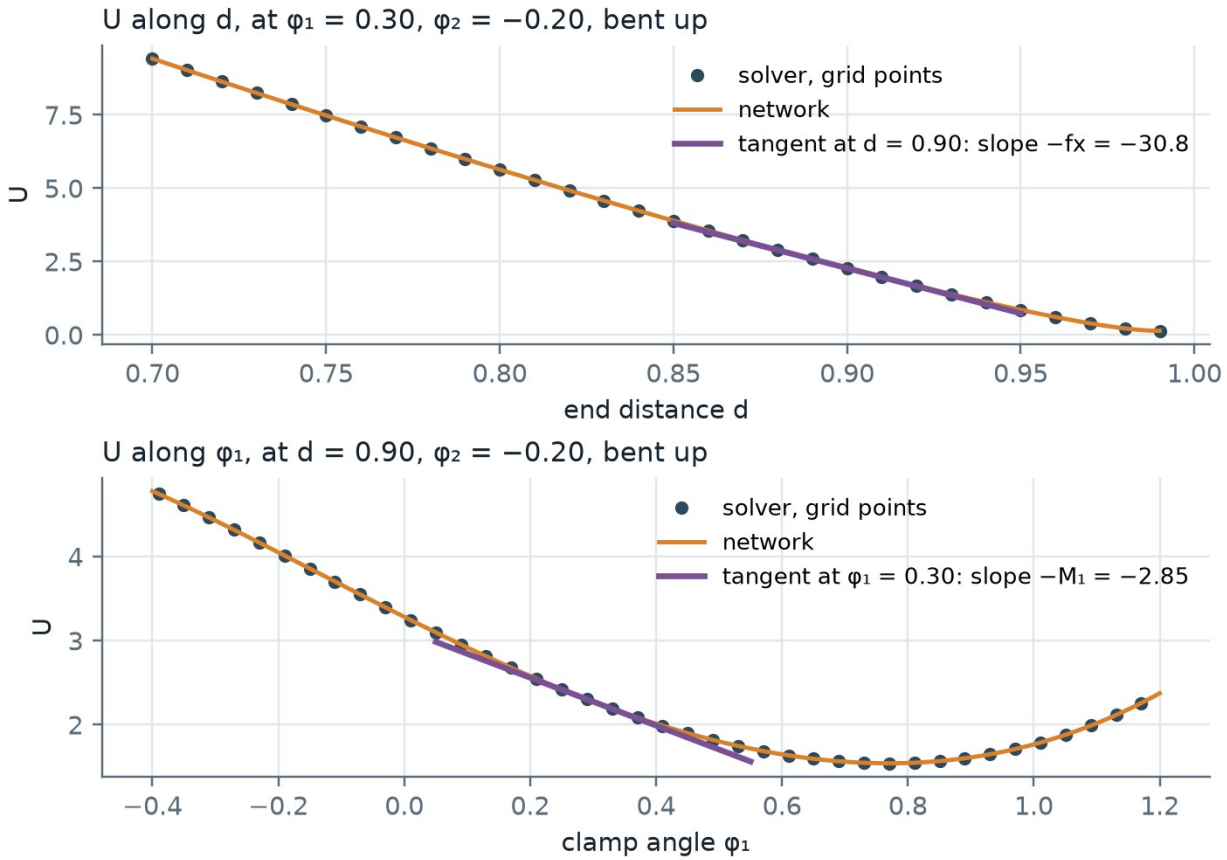
2 **The energy network**
Gives U, M_1 and M_2 of that shape. It works with ln U and asinh M rather than U and M (slides 57 and 58).

3 **The existence network**
Gives the probability that the shape exists there: past a fold the rod has snapped, and there is no such state.

4 **The forces**
Not separate outputs: fx and fy are slopes of the network's U, computed exactly (next slide).

Why can the forces come from the energy?

Because the end loads are the slopes of U: fx = −∂U/∂d, M_1 = −∂U/∂φ_1, M_2 = ∂U/∂φ_2, and fy follows from a rotation.



1 **Work done at the ends**
Moving the clamps changes the stored energy by the work of the end loads: dU = −fx dd − M_1 dφ_1 + M_2 dφ_2 (the envelope theorem).

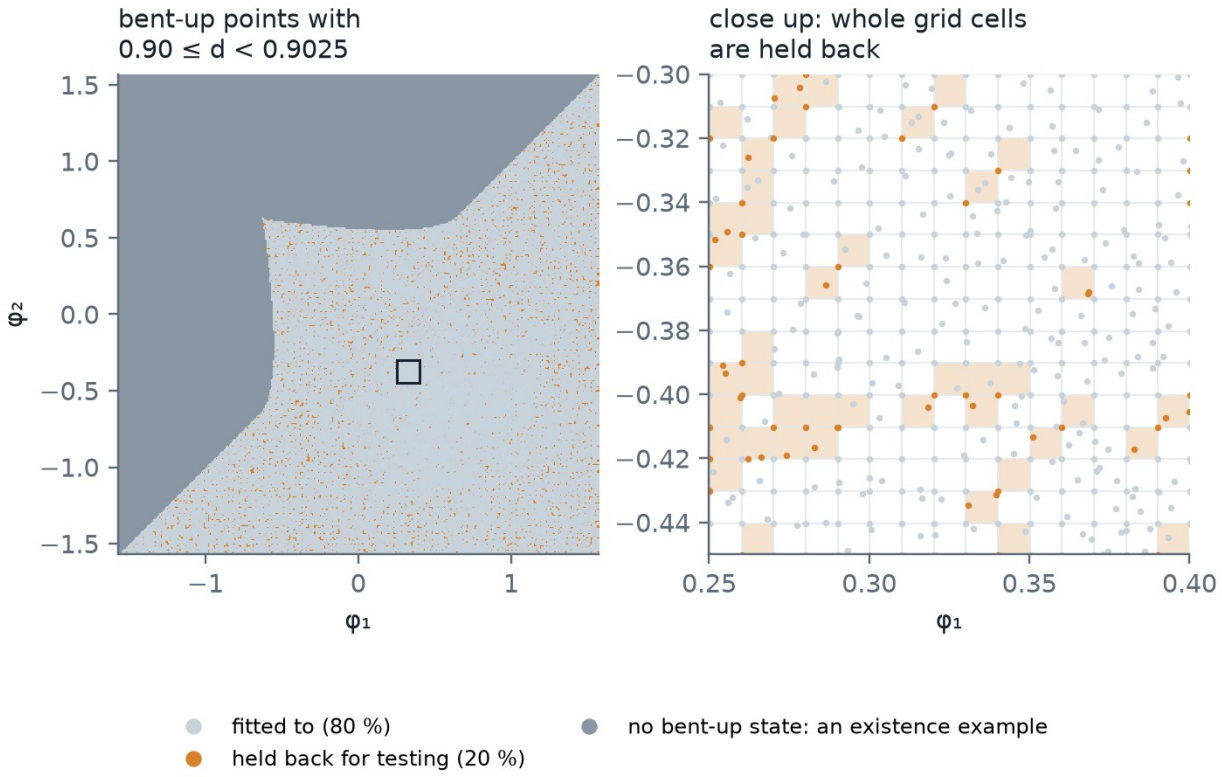
2 **fy from a rotation**
Turning the whole rod by α changes nothing but moves its end sideways by d α, so fy = (∂U/∂φ_1 + ∂U/∂φ_2) / d.

3 **Checked on the solver**
Slopes of the solver's own U between grid points match its fx, M_1 and M_2 to 1 part in 10^4. Left: the tangents, at one clamp setting.

4 **What it buys**
Forces consistent with the energy by construction, from exact derivatives of the formula, not from differences between nearby points.

Which data is it fitted to, and how is it tested?

11.4 million solver results. One grid cell in five is held back, whole, and used only to measure the error.



1 The states

9.04 million solved states of both shapes, denser where the energy is low (the refinement schemes earlier in the deck).

2 The no-state points

2.34 million grid points past a fold, where the chosen shape does not exist: the examples for the existence network.

3 Held back

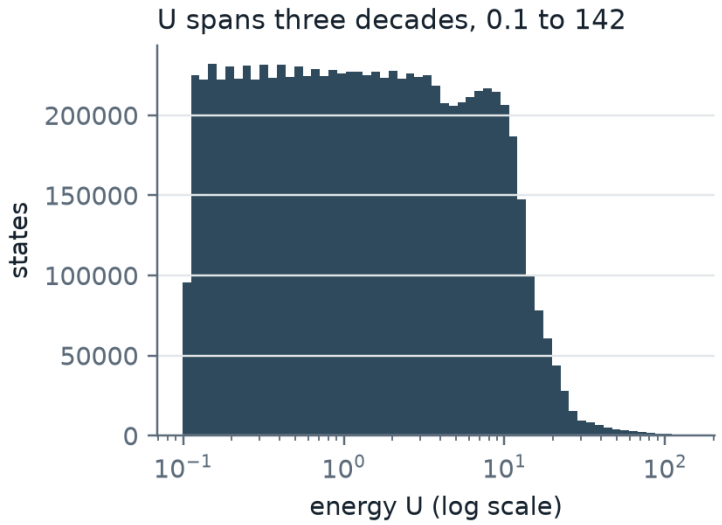
20 % of the grid cells, drawn at random in each energy band, are never shown to the fit. Every error on the following slides is measured on them.

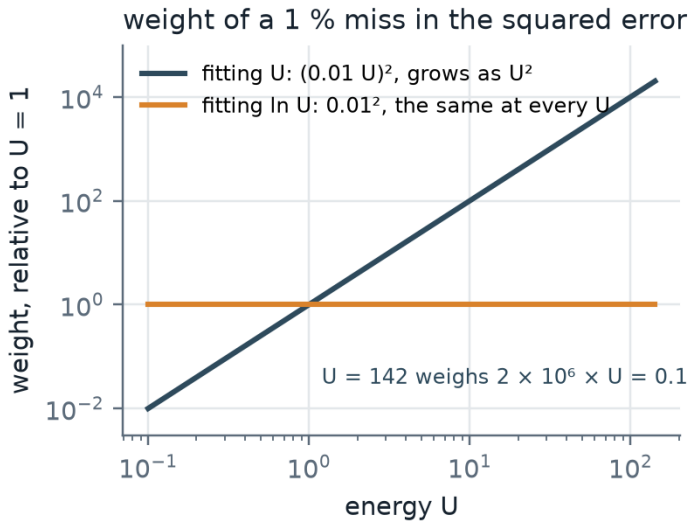
4 Mirror images together

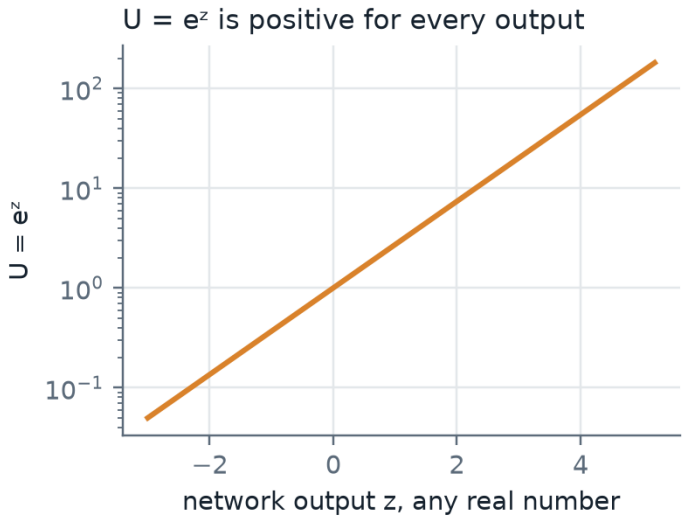
A state and its mirror images always fall on the same side, so no held-back state is the mirror of a fitted one.

Why fit ln U instead of U?

U spans 0.1 to 142. Fitted directly, the large energies would dominate the fit; fitted as ln U, every state counts by its relative error.







$$\ln(U+\delta U)-\ln U \approx \frac{\delta U}{U}$$

A miss in ln U is a relative miss in U: 0.001 in ln U is 0.1 % of U, at U = 0.1 and at U = 100 alike.

$$(\delta U)^2 = (0.01\ U)^2 \qquad \text{against} \qquad (\delta \ln U)^2 = 0.01^2$$

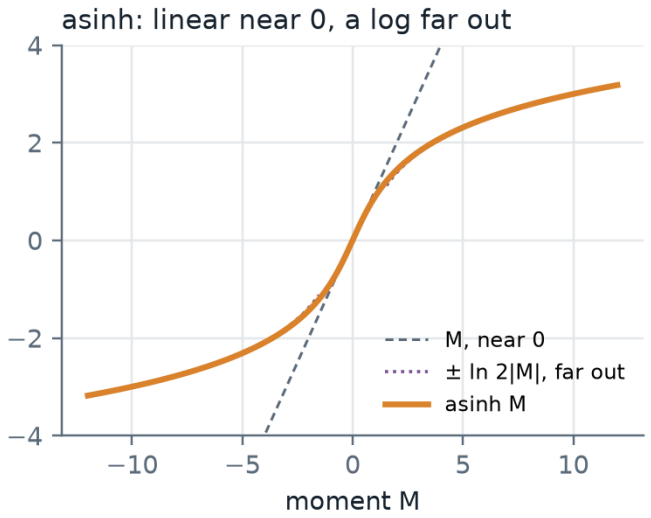
For the same 1 % miss, the plain squared error weighs a state at U = 142 two million times one at U = 0.1 (middle); ln U weighs them equally.

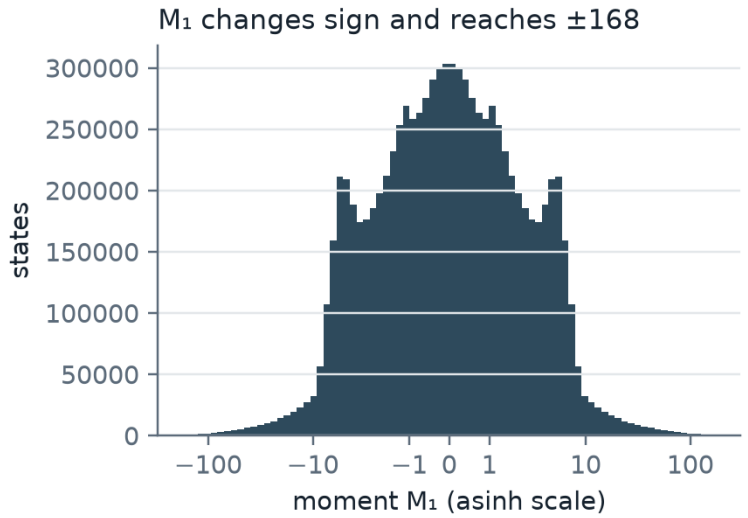
$$U = e^{z} > 0 \quad \text{for every } z$$

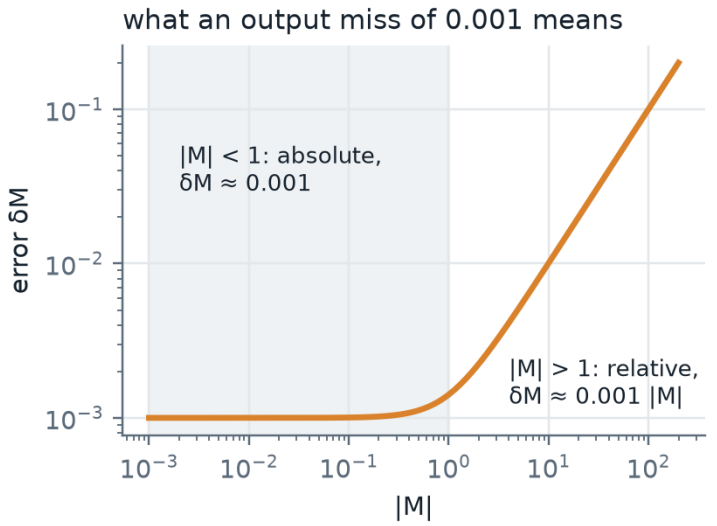
The network's output z can be any number, and U = eᶻ is still a positive energy (right).

Why asinh for the moments and forces?

They change sign, so ln is ruled out. asinh is linear near zero and logarithmic far from it: absolute accuracy near 0, relative when large.







$$\operatorname{asinh} M = \ln\left(M + \sqrt{M^2+1}\right)$$

Defined for every M, positive or negative. Used for M_1 and M_2, and for fx / 10 when the force is fitted.

$$\operatorname{asinh} M \approx M \quad (|M| \ll 1), \qquad \operatorname{asinh} M \approx \pm \ln 2|M| \quad (|M| \gg 1)$$

Its two limits (left): it follows M near zero, and the logarithm of |M| far from it.

$$\delta M \approx \sqrt{1+M^2}\ \delta(\operatorname{asinh} M), \qquad \operatorname{asinh}(-M) = -\operatorname{asinh} M$$

So a fixed miss is an absolute error below |M| = 1 and a relative one above (right). And it is odd: bent down, which mirrors M to −M, stays a mirror.

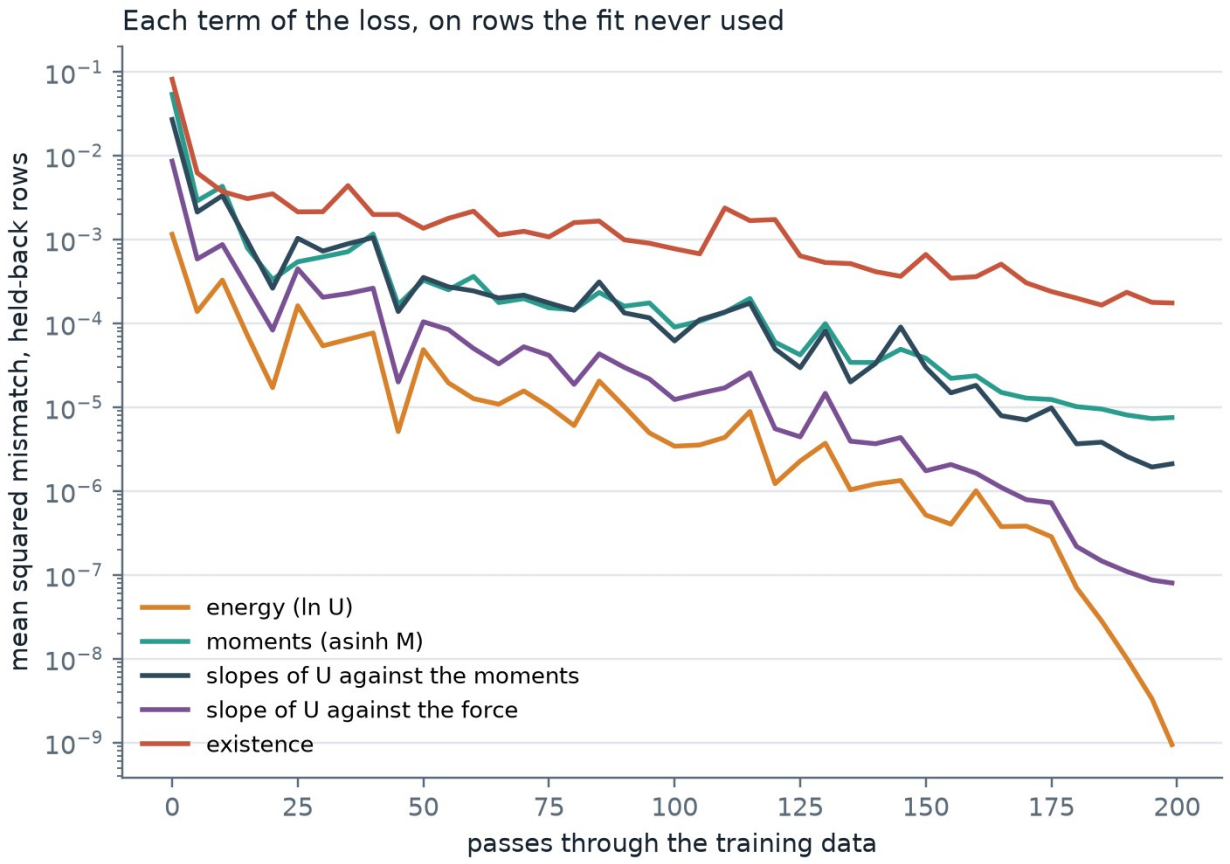
What exactly does the fit minimise?

One number, the loss: the average squared mismatch with the solver, summed over five terms. Fitting lowers it, batch after batch.

$\mathcal{L}_U = \left\langle (\ln U - \ln U^*)^2 \right\rangle$	Energy: ln U of the network against ln U* of the solver. ⟨ ⟩ is the average over the states in a batch; * marks the solver's value.
$\mathcal{L}_M = \left\langle \sum_{i=1,2} (\operatorname{asinh} M_i - \operatorname{asinh} M_i^*)^2 \right\rangle$	Moments, on the asinh scale of slide 58.
$\mathcal{L}_\partial = \left\langle \left(\operatorname{asinh} \frac{-\partial U}{\partial \varphi_1} - \operatorname{asinh} M_1^* \right)^2 + \left(\operatorname{asinh} \frac{\partial U}{\partial \varphi_2} - \operatorname{asinh} M_2^* \right)^2 \right\rangle$	The network's slopes of U against the solver's moments: makes the energy and the moments agree (slide 55).
$\mathcal{L}_f = \left\langle \left(\operatorname{asinh} \frac{-\partial U/\partial d}{f_0} - \operatorname{asinh} \frac{f_x^*}{f_0} \right)^2 \right\rangle, \quad f_0 = 10$	The network's slope along d against the solver's force (slide 65). f_0 = 10 sets where asinh turns from linear to logarithmic.
$\mathcal{L}_\exists = -\langle e \ln P + (1-e) \ln(1-P) \rangle$	Existence: e = 1 where the shape exists, else 0. The penalty grows without bound for a confident wrong answer.
$\mathcal{L} = \mathcal{L}_U + \mathcal{L}_M + \mathcal{L}_\partial + \mathcal{L}_f + \mathcal{L}_\exists$	Each term weighted 1. The energy network sees only states that exist; the existence network sees every point.

Does the fit converge without memorising?

Over 200 passes through the data every term falls by 2½ to 6 decades, and states the fit never saw come out as well as the ones it did.



1 **A pass**

Every training state used once, in batches of 8192; each batch gives one small correction to all the weights: 1,113 corrections per pass.

2 **Measured on held-back rows**

The curves are the loss terms on rows the fit never uses, every 5 passes. The early jumps come from large steps, which shrink towards the end.

3 **No memorising**

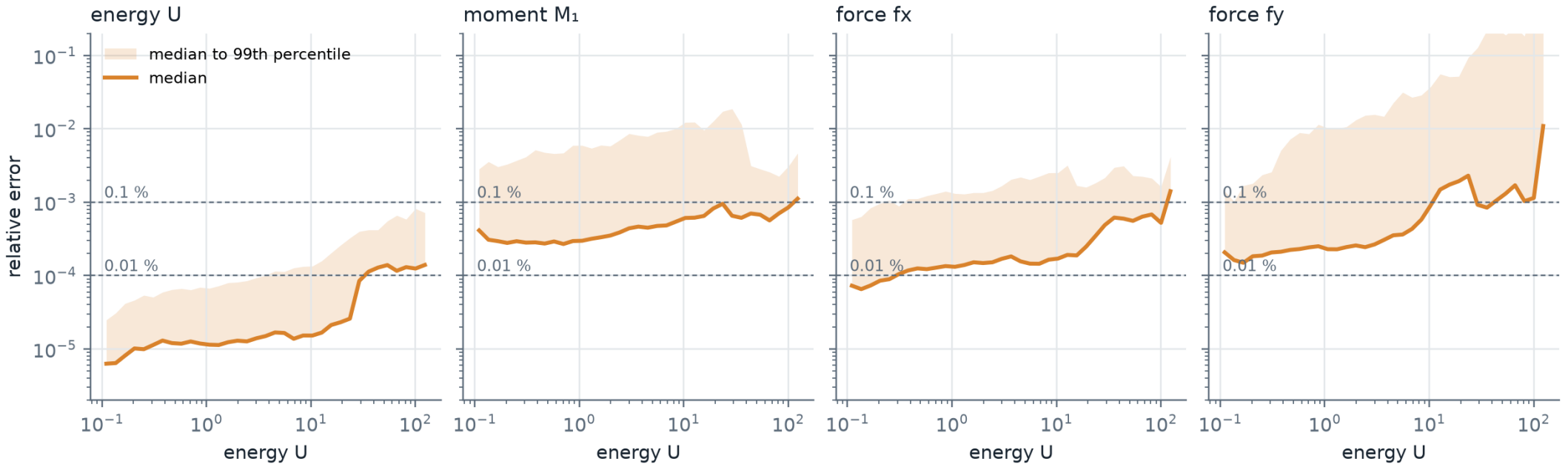
Median errors on fitted and on held-back states agree to two digits in every energy band (slide 66): the fit has not just stored its data.

4 **Cost**

86 minutes on one RTX A6000 graphics card, for 662 thousand weights and 11.4 million rows.

How close is it on states it never saw?

Half of the held-back states are within 0.0012 % in U and 0.04 % in the moments and forces.



Relative error = |network − solver| / |solver|, with a floor of 1 % of the typical size where a quantity passes through zero (the moments, fy); 1.8 million held-back states of both shapes, in 35 bins of energy. fy's 99th percentile is high only where fy itself is close to zero.

0.0012 %

energy U

median; 99 % within 0.011 %

0.036 %

moment M_1

median; 99 % within 0.61 %

0.013 %

force fx

median; 99 % within 0.16 %

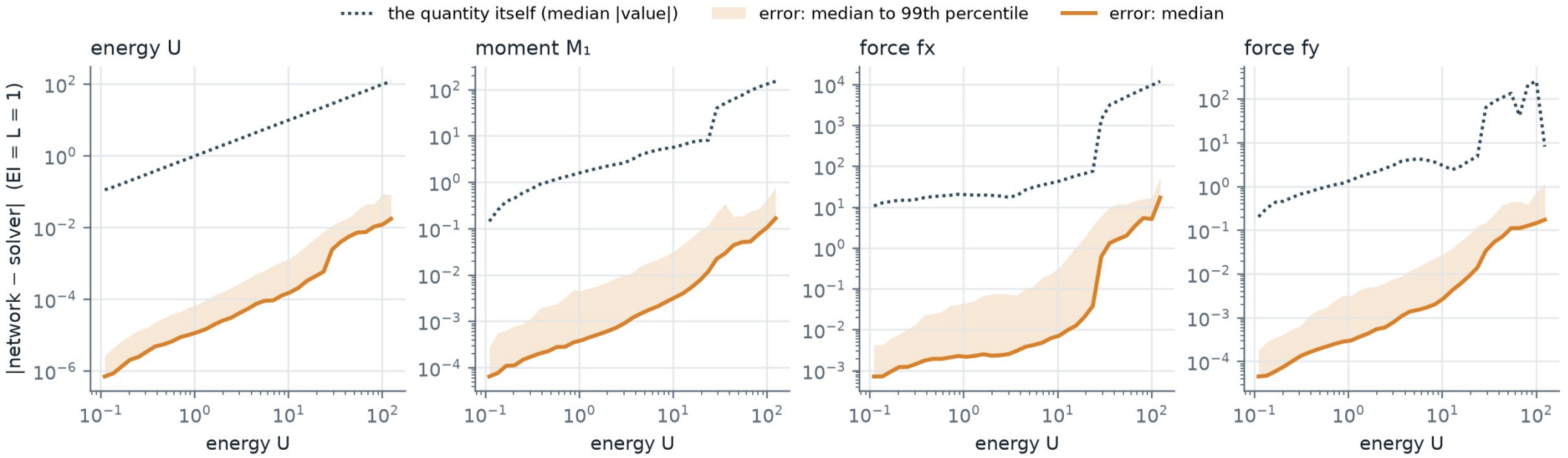
0.027 %

force fy

median; 99 % within 2 %

How big are the errors in the problem's units?

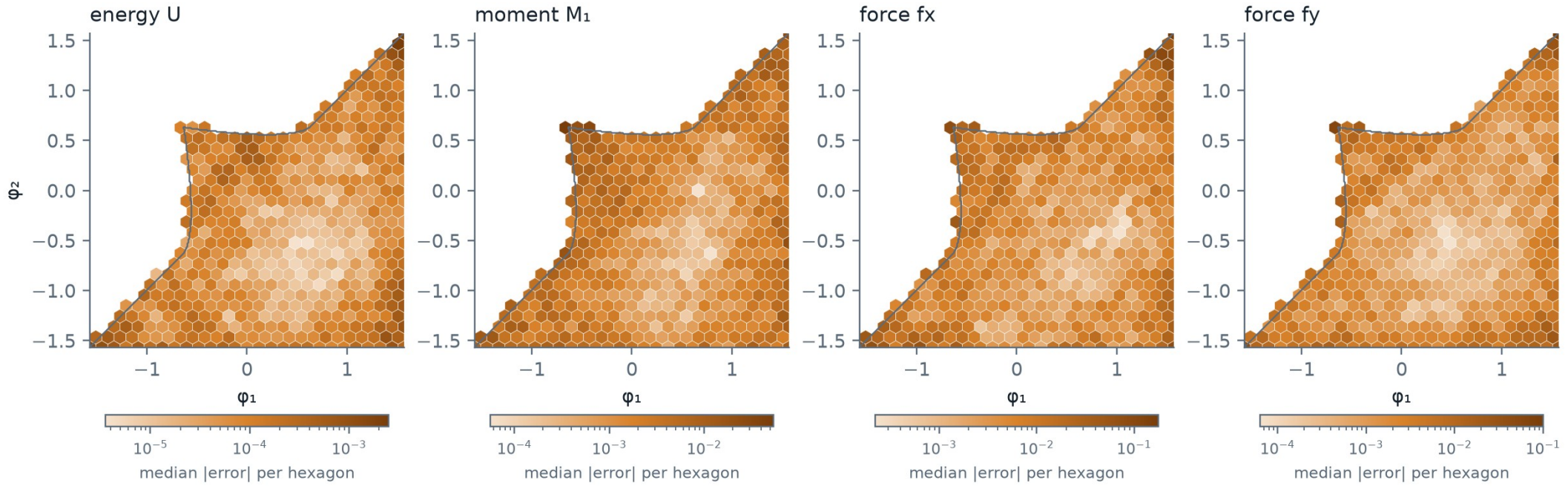
They grow with U about as fast as the quantities themselves, and stay a median 3½ to 5 decades below them.



EI = L = 1. Dotted: the typical size of the quantity at that energy; shaded: the error from its median to its 99th percentile, on the held-back states. U is off by 3×10^{-6} below U = 1 and by 2.6×10^{-4} above U = 10; the moments and fy by 1.5×10^{-4} to 5×10^{-3}; fx by 1.4×10^{-3} to 1.2×10^{-2}. Above U = 30, where |fx| reaches 1.4×10^{4}, fx is off by a median of 1.8.

Where are the errors largest?

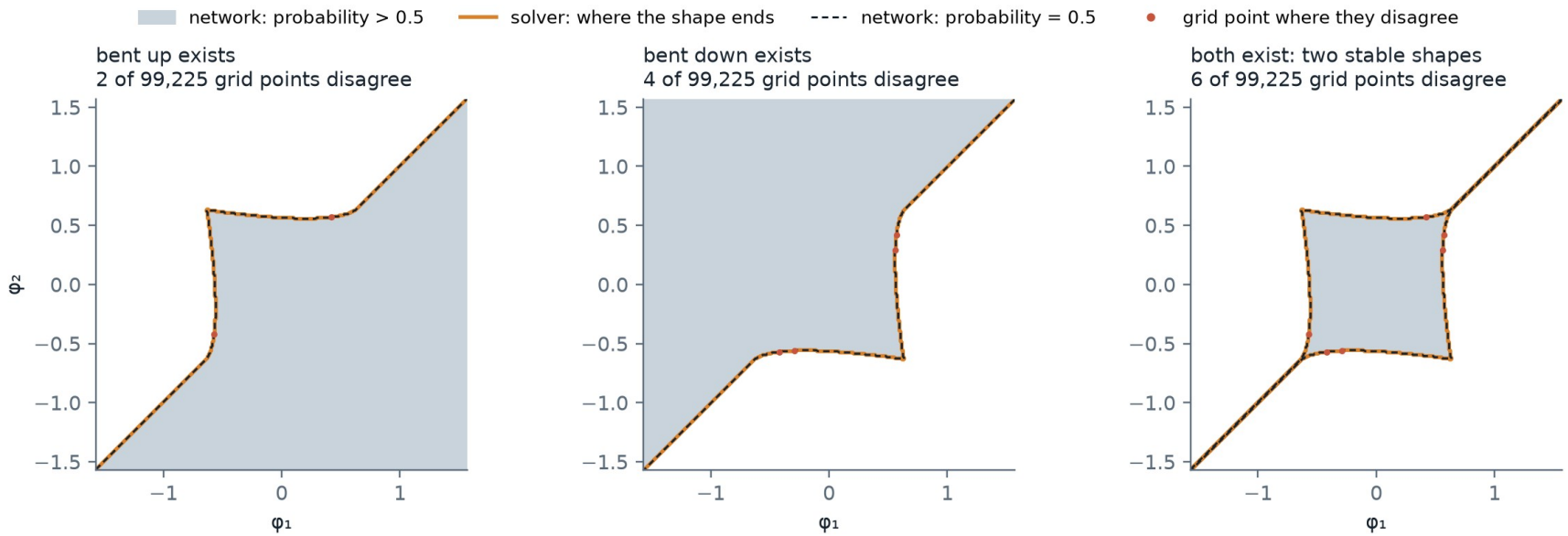
Along the fold, where a shape is about to snap: there the errors are 5 to 11 times those inside. The corners of the box come next.



Median |network − solver| per hexagon, over the 15,773 held-back bent-up states with |d − 0.90| < 0.0025; grey line: where the bent-up shape ends. Within 0.06 of the fold the median error is 1.5 × 10^{-4} in U against 2.3 × 10^{-5} inside, 6.5 × 10^{-3} against 6.0 × 10^{-4} in M_1, 1.3 × 10^{-2} against 2.4 × 10^{-3} in fx. The relative error is also 2 to 4 times higher there for M_1 and fx: near a fold the state changes fastest with the clamps.

Does it know where each shape exists?

Yes: 99.994 % of the held-back points are classified right, and the mistakes sit on the edges, within a grid step of where a shape ends.



d = 0.90, on the solver's grid. The edges are the folds, where a shape ends and the rod snaps, and the line $\varphi_1 = \varphi_2$, where the one state there turns from bent up to bent down.

99.994 %
held-back points right
2.27 M points, both shapes

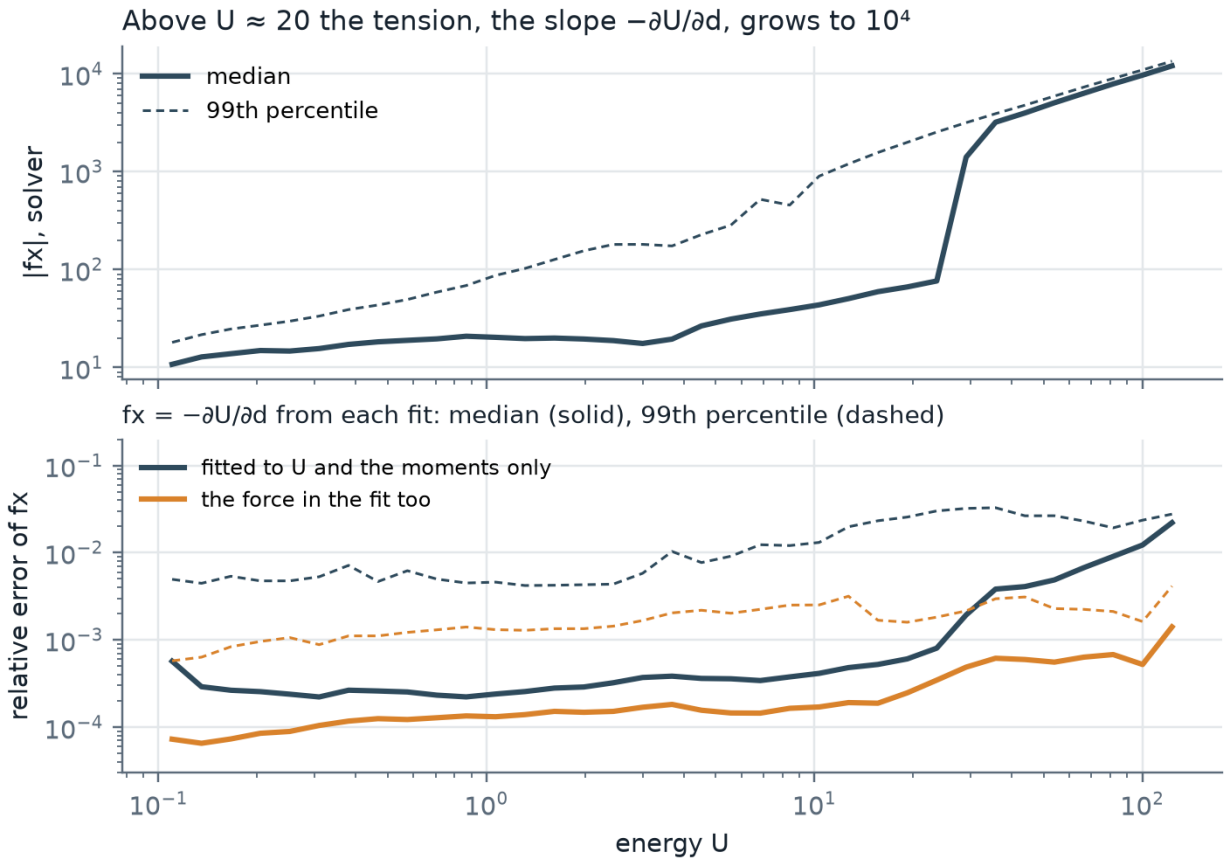
99.3 %
right in fold cells
16,572 points at an edge

0.004 %
shapes missed
exists, predicted absent

0.014 %
false shapes
absent, predicted to exist

Why does the fit use the solver's forces too?

Fitted on U and the moments alone, ∂U/∂d is off by up to 2 % at high tension; fitting the force too holds fx to 0.01–0.1 %.



1 **The tension grows steeply**

Above U ≈ 20, near d = 0.99 with large clamp angles, |fx| rises to 10^4 (top): there U changes very fast with d.

2 **A small slope error, a large force error**

A tiny error in the shape of U along d then becomes a large error in its slope, fx = −∂U/∂d (bottom, grey).

3 **Not a lack of data**

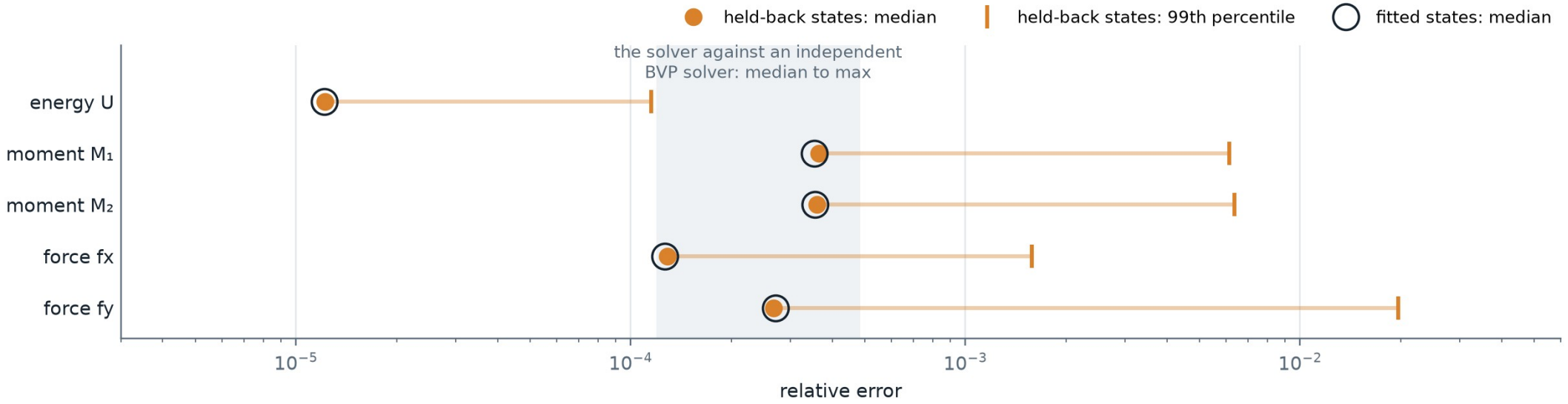
That fit is as wrong on the states it was fitted to as on new ones: the fit itself falls short, not the number of states.

4 **Fitting the force too**

The force term (slide 59) makes the slope match the solver's force: fx within 0.01–0.1 % at every energy, and U improves as well.

Is it good enough, and where does it stop?

Its errors are about the size of the solver's own, 1.2×10^{-4} against an independent solver: to do better, the solver has to improve first.



1 As accurate as its data

The solver agrees with an independent boundary-value solver to 1.2×10^{-4} (median); the network's moment and force errors are of that size (shaded band).

2 More data would not help

The errors on fitted states (rings) and on held-back ones (dots) are the same: the network is not short of data.

3 Where it holds

d from 0.70 to 0.99, φ_1 and φ_2 within $\pm\pi/2$, and only where the existence probability is above 0.5; least accurate along the folds.

4 To go further

A finer beam mesh near the clamps at high tension, then a larger network; more points near the folds for the existence network.